CHAPTER 2

SIMPLE LINEAR REGRESSION

2.1 SIMPLE LINEAR REGRESSION MODEL

This chapter considers the **simple linear regression model**, that is, a model with a single regressor x that has a relationship with a response y that is a straight line. This simple linear regression model is

$$y = \beta_0 + \beta_1 x + \varepsilon \tag{2.1}$$

where the intercept β_0 and the slope β_1 are unknown constants and ε is a random error component. The errors are assumed to have mean zero and unknown variance σ^2. Additionally we usually assume that the errors are uncorrelated. This means that the value of one error does not depend on the value of any other error.

It is convenient to view the regressor x as controlled by the data analyst and measured with negligible error, while the response y is a random variable. That is, there is a probability distribution for y at each possible value for x. The mean of this distribution is

$$E(y|x) = \beta_0 + \beta_1 x \tag{2.2a}$$

and the variance is

$$\mathrm{Var}(y|x) = \mathrm{Var}(\beta_0 + \beta_1 x + \varepsilon) = \sigma^2 \tag{2.2b}$$

Introduction to Linear Regression Analysis, Fifth Edition. Douglas C. Montgomery, Elizabeth A. Peck, G. Geoffrey Vining.

Thus, the mean of y is a linear function of x although the variance of y does not depend on the value of x. Furthermore, because the errors are uncorrelated, the responses are also uncorrelated.

The parameters β_0 and β_1 are usually called **regression coefficients**. These coefficients have a simple and often useful interpretation. The slope β_1 is the change in the mean of the distribution of y produced by a unit change in x. If the range of data on x includes $x = 0$, then the intercept β_0 is the mean of the distribution of the response y when $x = 0$. If the range of x does not include zero, then β_0 has no practical interpretation.

2.2 LEAST-SQUARES ESTIMATION OF THE PARAMETERS

The parameters β_0 and β_1 are unknown and must be estimated using sample data. Suppose that we have n pairs of data, say $(y_1, x_1), (y_2, x_2), \ldots, (y_n, x_n)$. As noted in Chapter 1, these data may result either from a controlled experiment designed specifically to collect the data, from an observational study, or from existing historical records (a retrospective study).

2.2.1 Estimation of β_0 and β_1

The **method of least squares** is used to estimate β_0 and β_1. That is, we estimate β_0 and β_1 so that the sum of the squares of the differences between the observations y_i and the straight line is a minimum. From Eq. (2.1) we may write

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \ldots, n \tag{2.3}$$

Equation (2.1) maybe viewed as a **population regression model** while Eq. (2.3) is a **sample regression model**, written in terms of the n pairs of data (y_i, x_i) $(i = 1, 2, \ldots, n)$. Thus, the least-squares criterion is

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i\right)^2 \tag{2.4}$$

The least-squares estimators of β_0 and β_1, say $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

$$\left.\frac{\partial S}{\partial \beta_0}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2\sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0$$

and

$$\left.\frac{\partial S}{\partial \beta_1}\right|_{\hat{\beta}_0, \hat{\beta}_1} = -2\sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) x_i = 0$$

Simplifying these two equations yields

$$
\begin{aligned}
n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i &= \sum_{i=1}^{n} y_i \\
\hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} y_i x_i
\end{aligned}
\tag{2.5}
$$

Equations (2.5) are called the **least-squares normal equations**. The solution to the normal equations is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{2.6}$$

and

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \dfrac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \dfrac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \tag{2.7}$$

where

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i \quad \text{and} \quad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

are the averages of y_i and x_i, respectively. Therefore, $\hat{\beta}_0$ and $\hat{\beta}_1$ in Eqs. (2.6) and (2.7) are the **least-squares estimators** of the intercept and slope, respectively. The fitted simple linear regression model is then

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \tag{2.8}$$

Equation (2.8) gives a point estimate of the mean of y for a particular x.

Since the denominator of Eq. (2.7) is the corrected sum of squares of the x_i and the numerator is the corrected sum of cross products of x_i and y_i, we may write these quantities in a more compact notation as

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} = \sum_{i=1}^{n} (x_i - \bar{x})^2 \tag{2.9}$$

and

$$S_{xy} = \sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n} = \sum_{i=1}^{n} y_i (x_i - \bar{x}) \tag{2.10}$$

Thus, a convenient way to write Eq. (2.7) is

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \tag{2.11}$$

The difference between the observed value y_i and the corresponding fitted value $\hat{y}_i$ is a **residual**. Mathematically the ith residual is

$$e_i = y_i - \hat{y}_i = y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right), \quad i = 1, 2, \ldots, n \tag{2.12}$$

Residuals play an important role in investigating **model adequacy** and in detecting departures from the underlying assumptions. This topic is discussed in subsequent chapters.

Example 2.1 The Rocket Propellant Data

A rocket motor is manufactured by bonding an igniter propellant and a sustainer propellant together inside a metal housing. The shear strength of the bond between the two types of propellant is an important quality characteristic. It is suspected that shear strength is related to the age in weeks of the batch of sustainer propellant. Twenty observations on shear strength and the age of the corresponding batch of propellant have been collected and are shown in Table 2.1. The scatter diagram, shown in Figure 2.1, suggests that there is a strong statistical relationship between shear strength and propellant age, and the tentative assumption of the straight-line model $y = \beta_0 + \beta_1 x + \varepsilon$ appears to be reasonable.

TABLE 2.1 Data for Example 2.1

Observation, i	Shear Strength, y_i (psi)	Age of Propellant, x_i (weeks)
1	2158.70	15.50
2	1678.15	23.75
3	2316.00	8.00
4	2061.30	17.00
5	2207.50	5.50
6	1708.30	19.00
7	1784.70	24.00
8	2575.00	2.50
9	2357.90	7.50
10	2256.70	11.00
11	2165.20	13.00
12	2399.55	3.75
13	1779.80	25.00
14	2336.75	9.75
15	1765.30	22.00
16	2053.50	18.00
17	2414.40	6.00
18	2200.50	12.50
19	2654.20	2.00
20	1753.70	21.50

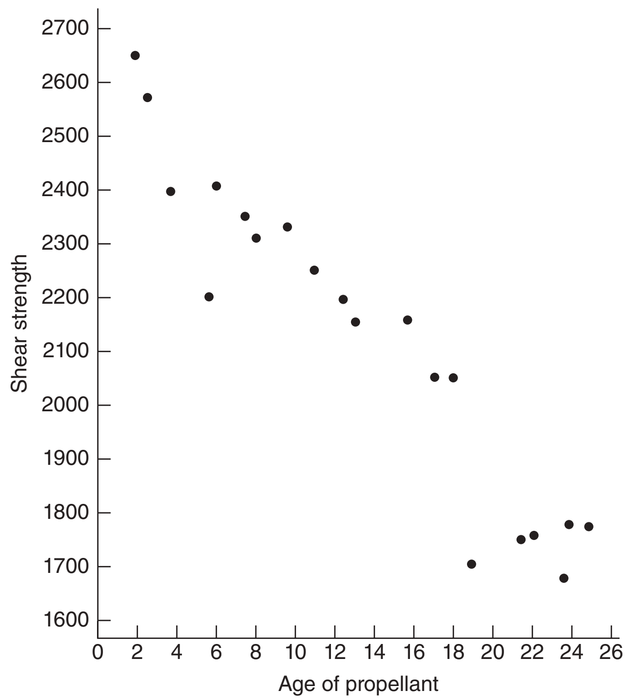


Figure 2.1 Scatter diagram of shear strength versus propellant age, Example 2.1.

To estimate the model parameters, first calculate

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} = 4677.69 - \frac{71,422.56}{20} = 1106.56$$

and

$$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} = 528,492.64 - \frac{(267.25)(42,627.15)}{20} = -41,112.65$$

Therefore, from Eqs. (2.11) and (2.6), we find that

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-41,112.65}{1106.56} = -37.15$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 2131.3575 - (-37.15)13.3625 = 2627.82$$

TABLE 2.2 Data, Fitted Values, and Residuals for Example 2.1

Observed Value, y_i	Fitted Value, $\hat{y}_i$	Residual, e_i
2158.70	2051.94	106.76
1678.15	1745.42	–67.27
2316.00	2330.59	–14.59
2061.30	1996.21	65.09
2207.50	2423.48	–215.98
1708.30	1921.90	–213.60
1784.70	1736.14	48.56
2575.00	2534.94	40.06
2357.90	2349.17	8.73
2256.70	2219.13	37.57
2165.20	2144.83	20.37
2399.55	2488.50	–88.95
1799.80	1698.98	80.82
2336.75	2265.58	71.17
1765.30	1810.44	–45.14
2053.50	1959.06	94.44
2414.40	2404.90	9.50
2200.50	2163.40	37.10
2654.20	2553.52	100.68
1753.70	1829.02	–75.32
$\sum y_i = 42,627.15$	$\sum \hat{y}_i = 42,627.15$	$\sum e_i = 0.00$

The least-squares fit is

$$\hat{y} = 2627.82 - 37.15x$$

We may interpret the slope –37.15 as the average weekly decrease in propellant shear strength due to the age of the propellant. Since the lower limit of the x's is near the origin, the intercept 2627.82 represents the shear strength in a batch of propellant immediately following manufacture. Table 2.2 displays the observed values y_i, the fitted values $\hat{y}_i$, and the residuals. ■

After obtaining the least-squares fit, a number of interesting questions come to mind:

1. How well does this equation fit the data?
2. Is the model likely to be useful as a predictor?
3. Are any of the basic assumptions (such as constant variance and uncorrelated errors) violated, and if so, how serious is this?

All of these issues must be investigated before the model is finally adopted for use. As noted previously, the residuals play a key role in evaluating model adequacy. Residuals can be viewed as realizations of the model errors ε_i. Thus, to check the constant variance and uncorrelated errors assumption, we must ask ourselves if the residuals look like a random sample from a distribution with these properties. We

TABLE 2.3 Minitab Regression Output for Example 2.1

```
Regression Analysis

The regression equation is
Strength = 2628- 37.2 Age

Predictor      Coef          StDev       T            P
Constant     2627.82         44.18      59.47        0.000
Age           -37.154         2.889    -12.86        0.000

S = 96.11    R-Sq = 90.2%   R-Sq(adj) = 89.6%

Analysis of Variance

Source          DF              SS        MS              F        P
Regression       1         1527483   1527483         165.38    0.000
Error           18          166255      9236
Total           19         1693738
```

return to these questions in Chapter 4, where the use of residuals in model adequacy checking is explored.

Computer Output Computer software packages are used extensively in fitting regression models. Regression routines are found in both network and PC-based statistical software, as well as in many popular spreadsheet packages. Table 2.3 presents the output from Minitab, a widely used PC-based statistics package, for the rocket propellant data in Example 2.1. The upper portion of the table contains the fitted regression model. Notice that before rounding the regression coefficients agree with those we calculated manually. Table 2.3 also contains other information about the regression model. We return to this output and explain these quantities in subsequent sections.

2.2.2 Properties of the Least-Squares Estimators and the Fitted Regression Model

The least-squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ have several important properties. First, note from Eqs. (2.6) and (2.7) that $\hat{\beta}_0$ and $\hat{\beta}_1$ are **linear combinations** of the observations y_i. For example,

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^{n} c_i y_i$$

where $c_i = (x_i - \bar{x})/S_{xx}$ for $i = 1, 2, \ldots, n$.

The least-squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are **unbiased estimators** of the model parameters β_0 and β_1. To show this for $\hat{\beta}_1$, consider

$$E\left(\hat{\beta}_1\right) = E\left(\sum_{i=1}^{n} c_i y_i\right) = \sum_{i=1}^{n} c_i E(y_i)$$

$$= \sum_{i=1}^{n} c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^{n} c_i + \beta_1 \sum_{i=1}^{n} c_i x_i$$

since $E(\varepsilon_i) = 0$ by assumption. Now we can show directly that $\sum_{i=1}^{n} c_i = 0$ and $\sum_{i=1}^{n} c_i x_i = 1$, so

$$E(\hat{\beta}_1) = \beta_1$$

That is, if we assume that the model is correct $[E(y_i) = \beta_0 + \beta_1 x_i]$, then $\hat{\beta}_1$ is an unbiased estimator of β_1. Similarly we may show that $\hat{\beta}_0$ is an unbiased estimator of β_0, or

$$E(\hat{\beta}_0) = \beta_0$$

The variance of $\hat{\beta}_1$ is found as

$$\operatorname{Var}(\hat{\beta}_1) = \operatorname{Var}\left(\sum_{i=1}^{n} c_i y_i\right) = \sum_{i=1}^{n} c_i^2 \operatorname{Var}(y_i) \tag{2.13}$$

because the observations y_i are uncorrelated, and so the variance of the sum is just the sum of the variances. The variance of each term in the sum is $c_i^2 \operatorname{Var}(y_i)$, and we have assumed that $\operatorname{Var}(y_i) = \sigma^2$; consequently,

$$\operatorname{Var}(\hat{\beta}_1) = \sigma^2 \sum_{i=1}^{n} c_i^2 = \frac{\sigma^2 \sum_{i=1}^{n} (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}} \tag{2.14}$$

The variance of $\hat{\beta}_0$ is

$$\begin{aligned}
\operatorname{Var}(\hat{\beta}_0) &= \operatorname{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) \\
&= \operatorname{Var}(\bar{y}) + \bar{x}^2 \operatorname{Var}(\hat{\beta}_1) - 2\bar{x} \operatorname{Cov}(\bar{y}, \hat{\beta}_1)
\end{aligned}$$

Now the variance of $\bar{y}$ is just $\operatorname{Var}(\bar{y}) = \sigma^2/n$, and the covariance between $\bar{y}$ and $\hat{\beta}_1$ can be shown to be zero (see Problem 2.25). Thus,

$$\operatorname{Var}(\hat{\beta}_0) = \operatorname{Var}(\bar{y}) + \bar{x}^2 \operatorname{Var}(\hat{\beta}_1) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right) \tag{2.15}$$

Another important result concerning the quality of the least-squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ is the **Gauss-Markov theorem**, which states that for the regression model (2.1) with the assumptions $E(\varepsilon) = 0$, $\operatorname{Var}(\varepsilon) = \sigma^2$, and uncorrelated errors, the least-squares estimators are unbiased and have minimum variance when compared with all other unbiased estimators that are linear combinations of the y_i. We often say that the least-squares estimators are **best linear unbiased estimators**, where "best" implies minimum variance. Appendix C.4 proves the Gauss-Markov theorem for the more general multiple linear regression situation, of which simple linear regression is a special case.

There are several other useful properties of the least-squares fit:

1. The sum of the residuals in any regression model that contains an intercept β_0 is always zero, that is,

$$\sum_{i=1}^{n}(y_i - \hat{y}_i) = \sum_{i=1}^{n} e_i = 0$$

   This property follows directly from the first normal equation in Eqs. (2.5) and is demonstrated in Table 2.2 for the residuals from Example 2.1. Rounding errors may affect the sum.
2. The sum of the observed values y_i equals the sum of the fitted values $\hat{y}_i$, or

$$\sum_{i=1}^{n} y_i = \sum_{i=1}^{n} \hat{y}_i$$

   Table 2.2 demonstrates this result for Example 2.1.
3. The least-squares regression line always passes through the **centroid** [the point $(\bar{y}, \bar{x})$] of the data.
4. The sum of the residuals weighted by the corresponding value of the regressor variable always equals zero, that is,

$$\sum_{i=1}^{n} x_i e_i = 0$$

5. The sum of the residuals weighted by the corresponding fitted value always equals zero, that is,

$$\sum_{i=1}^{n} \hat{y}_i e_i = 0$$

2.2.3 Estimation of σ^2

In addition to estimating β_0 and β_1, an estimate of σ^2 is required to test hypotheses and construct interval estimates pertinent to the regression model. Ideally we would like this estimate not to depend on the adequacy of the fitted model. This is only possible when there are several observations on y for at least one value of x (see Section 4.5) or when prior information concerning σ^2 is available. When this approach cannot be used, the estimate of σ^2 is obtained from the **residual** or **error sum of squares**,

$$SS_{\text{Res}} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2.16}$$

A convenient computing formula for SS_{Res} may be found by substituting $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ into Eq. (2.16) and simplifying, yielding

$$SS_{\text{Res}} = \sum_{i=1}^{n} y_i^2 - n\bar{y}^2 - \hat{\beta}_1 S_{xy} \tag{2.17}$$

But

$$\sum_{i=1}^{n} y_i^2 - n\bar{y}^2 = \sum_{i=1}^{n} (y_i - \bar{y})^2 \equiv SS_{\text{T}}$$

is just the corrected sum of squares of the response observations, so

$$SS_{\text{Res}} = SS_{\text{T}} - \hat{\beta}_1 S_{xy} \tag{2.18}$$

The residual sum of squares has $n - 2$ degrees of freedom, because two degrees of freedom are associated with the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ involved in obtaining $\hat{y}_i$. Section C.3 shows that the expected value of SS_{Res} is $E(SS_{\text{Res}}) = (n - 2)\sigma^2$, so an **unbiased estimator of** σ^2 is

$$\hat{\sigma}^2 = \frac{SS_{\text{Res}}}{n-2} = MS_{\text{Res}} \tag{2.19}$$

The quantity MS_{Res} is called the **residual mean square**. The square root of $\hat{\sigma}^2$ is sometimes called the **standard error of regression**, and it has the same units as the response variable y.

Because $\hat{\sigma}^2$ depends on the residual sum of squares, any violation of the assumptions on the model errors or any misspecification of the model form may seriously damage the usefulness of $\hat{\sigma}^2$ as an estimate of σ^2. Because $\hat{\sigma}^2$ is computed from the regression model residuals, we say that it is a **model-dependent** estimate of σ^2.

Example 2.2 The Rocket Propellant Data

To estimate σ^2 for the rocket propellant data in Example 2.1, first find

$$\begin{aligned} SS_{\text{T}} &= \sum_{i=1}^{n} y_i^2 - n\bar{y}^2 = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n} \\ &= 92,547,433.45 - \frac{(42,627.15)^2}{20} = 1,693,737.60 \end{aligned}$$

From Eq. (2.18) the residual sum of squares is

$$\begin{aligned} SS_{\text{Res}} &= SS_{\text{T}} - \hat{\beta}_1 S_{xy} \\ &= 1,693,737.60 - (-37.15)(-41,112.65) = 166,402.65 \end{aligned}$$

Therefore, the estimate of σ^2 is computed from Eq. (2.19) as

$$\hat{\sigma}^2 = \frac{SS_{\text{Res}}}{n-2} = \frac{166,402.65}{18} = 9244.59$$

Remember that this estimate of σ^2 is **model dependent**. Note that this differs slightly from the value given in the Minitab output (Table 2.3) because of rounding. ■

2.2.4 Alternate Form of the Model

There is an alternate form of the simple linear regression model that is occasionally useful. Suppose that we redefine the regressor variable x_i as the deviation from its own average, say $x_i - \bar{x}$. The regression model then becomes

$$\begin{aligned} y_i &= \beta_0 + \beta_1(x_i - \bar{x}) + \beta_1\bar{x} + \varepsilon_i \\ &= (\beta_0 + \beta_1\bar{x}) + \beta_1(x_i - \bar{x}) + \varepsilon_i \\ &= \beta_0' + \beta_1(x_i - \bar{x}) + \varepsilon_i \end{aligned} \tag{2.20}$$

Note that redefining the regressor variable in Eq. (2.20) has shifted the origin of the x's from zero to $\bar{x}$. In order to keep the fitted values the same in both the original and transformed models, it is necessary to modify the original intercept. The relationship between the original and transformed intercept is

$$\beta_0' = \beta_0 + \beta_1\bar{x} \tag{2.21}$$

It is easy to show that the least-squares estimator of the transformed intercept is $\hat{\beta}_0' = \bar{y}$. The estimator of the slope is unaffected by the transformation. This alternate form of the model has some advantages. First, the least-squares estimators $\hat{\beta}_0' = \bar{y}$ and $\hat{\beta}_1 = S_{xy}/S_{xx}$ are **uncorrelated**, that is, $\mathrm{Cov}\left(\hat{\beta}_0', \hat{\beta}_1\right) = 0$. This will make some applications of the model easier, such as finding confidence intervals on the mean of y (see Section 2.4.2). Finally, the fitted model is

$$\hat{y} = \bar{y} + \hat{\beta}_1(x - \bar{x}) \tag{2.22}$$

Although Eqs. (2.22) and (2.8) are equivalent (they both produce the same value of $\hat{y}$ for the same value of x), Eq. (2.22) directly reminds the analyst that the regression model is only valid over the range of x in the **original data**. This region is centered at $\bar{x}$.

2.3 HYPOTHESIS TESTING ON THE SLOPE AND INTERCEPT

We are often interested in testing hypotheses and constructing confidence intervals about the model parameters. Hypothesis testing is discussed in this section, and Section 2.4 deals with confidence intervals. These procedures require that we make the additional assumption that the model errors ε_i are normally distributed. Thus, the complete assumptions are that the errors are normally and independently distributed with mean 0 and variance σ^2, abbreviated NID(0, σ^2). In Chapter 4 we discuss how these assumptions can be checked through **residual analysis**.

2.3.1 Use of *t* Tests

Suppose that we wish to test the hypothesis that the slope equals a constant, say β_{10}. The appropriate hypotheses are

$$H_0\colon \beta_1 = \beta_{10}, \quad H_1\colon \beta_1 \neq \beta_{10} \tag{2.23}$$

where we have specified a two-sided alternative. Since the errors ε_i are NID(0, σ^2), the observations y_i are NID($\beta_0 + \beta_1 x_i$, σ^2). Now $\hat{\beta}_1$ is a linear combination of the observations, so $\hat{\beta}_1$ is normally distributed with mean β_1 and variance σ^2/S_{xx} using the mean and variance of $\hat{\beta}_1$ found in Section 2.2.2. Therefore, the statistic

$$Z_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{\sigma^2/S_{xx}}}$$

is distributed $N(0, 1)$ if the null hypothesis H_0: $\beta_1 = \beta_{10}$ is true. If σ^2 were known, we could use Z_0 to test the hypotheses (2.23). Typically, σ^2 is unknown. We have already seen that MS_{Res} is an unbiased estimator of σ^2. Appendix C.3 establishes that $(n - 2)MS_{\text{Res}}/\sigma^2$ follows a χ^2_{n-2} distribution and that MS_{Res} and $\hat{\beta}_1$ are independent. By the definition of a t statistic given in Section C.1,

$$t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{MS_{\text{Res}}/S_{xx}}} \tag{2.24}$$

follows a t_{n-2} distribution if the null hypothesis H_0: $\beta_1 = \beta_{10}$ is true. The degrees of freedom associated with t_0 are the number of degrees of freedom associated with MS_{Res}. Thus, the ratio t_0 is the test statistic used to test H_0: $\beta_1 = \beta_{10}$. The test procedure computes t_0 and compares the observed value of t_0 from Eq. (2.24) with the upper $\alpha/2$ percentage point of the t_{n-2} distribution ($t_{\alpha/2,n-2}$). This procedure rejects the null hypothesis if

$$|t_0| > t_{\alpha/2,n-2} \tag{2.25}$$

Alternatively, a P-value approach could also be used for decision making.

The denominator of the test statistic, t_0, in Eq. (2.24) is often called the **estimated standard error**, or more simply, the **standard error** of the slope. That is,

$$\text{se}\left(\hat{\beta}_1\right) = \sqrt{\frac{MS_{\text{Res}}}{S_{xx}}} \tag{2.26}$$

Therefore, we often see t_0 written as

$$t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\text{se}\left(\hat{\beta}_1\right)} \tag{2.27}$$

A similar procedure can be used to test hypotheses about the intercept. To test

$$H_0\colon \beta_0 = \beta_{00}, \quad H_1\colon \beta_0 \neq \beta_{00} \tag{2.28}$$

we would use the **test statistic**

$$t_0 = \frac{\hat{\beta}_0 - \beta_{00}}{\sqrt{MS_{\text{Res}}\left(1/n + \bar{x}^2/S_{xx}\right)}} = \frac{\hat{\beta}_0 - \beta_{00}}{\text{se}\left(\hat{\beta}_0\right)} \tag{2.29}$$

where $\text{se}(\hat{\beta}_0) = \sqrt{MS_{\text{Res}}(1/n + \bar{x}^2/S_{xx})}$ is the **standard error of the intercept**. We reject the null hypothesis H_0: $\beta_0 = \beta_{00}$ if $|t_0| > t_{\alpha/2,n-2}$.

2.3.2 Testing Significance of Regression

A very important special case of the hypotheses in Eq. (2.23) is

$$H_0\text{: } \beta_1 = 0, \quad H_1\text{: } \beta_1 \neq 0 \tag{2.30}$$

These hypotheses relate to the **significance of regression**. Failing to reject H_0: $\beta_1 = 0$ implies that there is no linear relationship between x and y. This situation is illustrated in Figure 2.2. Note that this may imply either that x is of little value in explaining the variation in y and that the best estimator of y for any x is $\hat{y} = \bar{y}$ (Figure 2.2*a*) or that the true relationship between x and y is not linear (Figure 2.2*b*). Therefore, failing to reject H_0: $\beta_1 = 0$ is equivalent to saying that there is **no linear relationship between y and x**.

Alternatively, if H_0: $\beta_1 = 0$ is rejected, this implies that x is of value in explaining the variability in y. This is illustrated in Figure 2.3. However, rejecting H_0: $\beta_1 = 0$ could mean either that the straight-line model is adequate (Figure 2.3*a*) or that even

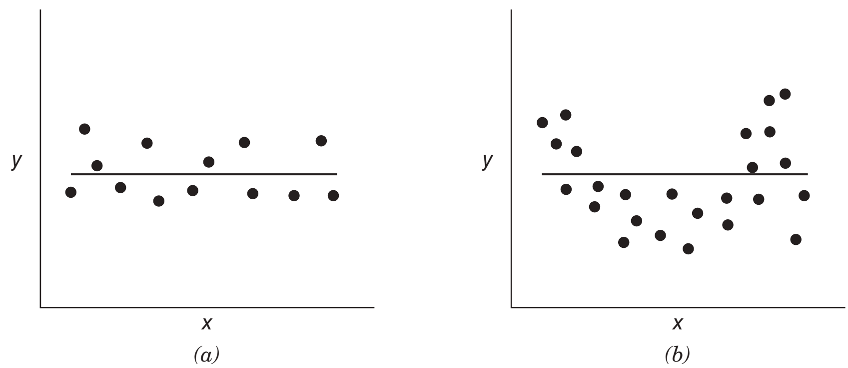


Figure 2.2 Situations where the hypothesis H_0: $\beta_1 = 0$ is not rejected.

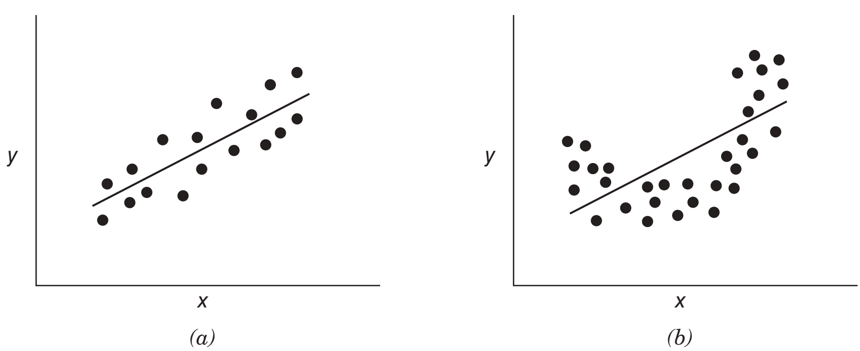


Figure 2.3 Situations where the hypothesis H_0: $\beta_1 = 0$ is rejected.

though there is a linear effect of x, better results could be obtained with the addition of higher order polynomial terms in x (Figure 2.3b).

The test procedure for H_0: $\beta_1 = 0$ may be developed from two approaches. The first approach simply makes use of the t statistic in Eq. (2.27) with $\beta_{10} = 0$, or

$$t_0 = \frac{\hat{\beta}_1}{\text{se}\left(\hat{\beta}_1\right)}$$

The null hypothesis of significance of regression would be rejected if $|t_0| > t_{\alpha/2,n-2}$.

Example 2.3 The Rocket Propellant Data

We test for significance of regression in the rocket propellant regression model of Example 2.1. The estimate of the slope is $\hat{\beta}_1 = -37.15$, and in Example 2.2, we computed the estimate of σ^2 to be $MS_{\text{Res}} = \hat{\sigma}^2 = 9244.59$. The standard error of the slope is

$$\text{se}\left(\hat{\beta}_1\right) = \sqrt{\frac{MS_{\text{Res}}}{S_{xx}}} = \sqrt{\frac{9244.59}{1106.56}} = 2.89$$

Therefore, the test statistic is

$$t_0 = \frac{\hat{\beta}_1}{\text{se}\left(\hat{\beta}_1\right)} = \frac{-37.15}{2.89} = -12.85$$

If we choose $\alpha = 0.05$, the critical value of t is $t_{0.025,18} = 2.101$. Thus, we would reject H_0: $\beta_1 = 0$ and conclude that there is a linear relationship between shear strength and the age of the propellant. ■

Minitab Output The Minitab output in Table 2.3 gives the standard errors of the slope and intercept (called "StDev" in the table) along with the t statistic for testing H_0: $\beta_1 = 0$ and H_0: $\beta_0 = 0$. Notice that the results shown in this table for the slope essentially agree with the manual calculations in Example 2.3. Like most computer software, Minitab uses the P-value approach to hypothesis testing. The P value for the test for significance of regression is reported as $P = 0.000$ (this is a rounded value; the actual P value is 1.64×10^{-10}). Clearly there is strong evidence that strength is linearly related to the age of the propellant. The test statistic for H_0: $\beta_0 = 0$ is reported as $t_0 = 59.47$ with $P = 0.000$. One would feel very confident in claiming that the intercept is not zero in this model.

2.3.3 Analysis of Variance

We may also use an **analysis-of-variance** approach to test significance of regression. The analysis of variance is based on a partitioning of total variability in the response variable y. To obtain this partitioning, begin with the identity

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i) \tag{2.31}$$

Squaring both sides of Eq. (2.31) and summing over all n observations produces

$$\sum_{i=1}^{n}(y_i-\bar{y})^2=\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2+2\sum_{i=1}^{n}(\hat{y}_i-\bar{y})(y_i-\hat{y}_i)$$

Note that the third term on the right-hand side of this expression can be rewritten as

$$\begin{aligned}2\sum_{i=1}^{n}(\hat{y}_i-\bar{y})(y_i-\hat{y}_i)&=2\sum_{i=1}^{n}\hat{y}_i(y_i-\hat{y}_i)-2\bar{y}\sum_{i=1}^{n}(y_i-\hat{y}_i)\\&=2\sum_{i=1}^{n}\hat{y}_ie_i-2y\sum_{i=1}^{n}e_i=0\end{aligned}$$

since the sum of the residuals is always zero (property 1, Section 2.2.2) and the sum of the residuals weighted by the corresponding fitted value $\hat{y}_i$ is also zero (property 5, Section 2.2.2). Therefore,

$$\sum_{i=1}^{n}(y_i-\bar{y})^2=\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2+\sum_{i=1}^{n}(y_i-\hat{y}_i)^2 \tag{2.32}$$

The left-hand side of Eq. (2.32) is the corrected sum of squares of the observations, SS_T, which measures the total variability in the observations. The two components of SS_T measure, respectively, the amount of variability in the observations y_i accounted for by the regression line and the residual variation left unexplained by the regression line. We recognize $SS_{Res}=\sum_{i=1}^{n}(y_i-\hat{y}_i)^2$ as the residual or error sum of squares from Eq. (2.16). It is customary to call $\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2$ the **regression** or **model sum of squares**.

Equation (2.32) is the **fundamental analysis-of-variance identity for a regression model**. Symbolically, we usually write

$$SS_T=SS_R+SS_{Res} \tag{2.33}$$

Comparing Eq. (2.33) with Eq. (2.18) we see that the regression sum of squares may be computed as

$$SS_R=\hat{\beta}_1S_{xy} \tag{2.34}$$

The **degree-of-freedom** breakdown is determined as follows. The total sum of squares, SS_T, has $df_T=n-1$ degrees of freedom because one degree of freedom is lost as a result of the constraint $\sum_{i=1}^{n}(y_i-\bar{y})$ on the deviations $y_i-\bar{y}$. The model or regression sum of squares, SS_R, has $df_R=1$ degree of freedom because SS_R is completely determined by one parameter, namely, $\hat{\beta}_1$ [see Eq. (2.34)]. Finally, we noted previously that SS_R has $df_{Res}=n-2$ degrees of freedom because two constraints are imposed on the deviations $y_i - \hat{y}_i$ as a result of estimating $\hat{\beta}_0$ and $\hat{\beta}_1$. Note that the degrees of freedom have an additive property:

$$\begin{aligned} df_T &= df_R + df_{Res} \\ n-1 &= 1+(n-2) \end{aligned} \tag{2.35}$$

We can use the usual **analysis-of-variance** F test to test the hypothesis H_0: $\beta_1 = 0$. Appendix C.3 shows that (1) $SS_{Res} = (n-2)MS_{Res}/\sigma^2$ follows a χ^2_{n-2} distribution; (2) if the null hypothesis H_0: $\beta_1 = 0$ is true, then SS_R/σ^2 follows a χ^2_1 distribution; and (3) SS_{Res} and SS_R are independent. By the definition of an F statistic given in Appendix C.1,

$$F_0 = \frac{SS_R/df_R}{SS_{Res}/df_{Res}} = \frac{SS_R/1}{SS_{Res}/(n-2)} = \frac{MS_R}{MS_{Res}} \tag{2.36}$$

follows the $F_{1,n-2}$ distribution. Appendix C.3 also shows that the expected values of these mean squares are

$$E(MS_{Res}) = \sigma^2, \quad E(MS_R) = \sigma^2 + \beta_1^2 S_{xx}$$

These expected mean squares indicate that if the observed value of F_0 is large, then it is likely that the slope $\beta_1 \neq 0$. Appendix C.3 also shows that if $\beta_1 \neq 0$, then F_0 follows a noncentral F distribution with 1 and $n-2$ degrees of freedom and a **noncentrality** parameter of

$$\lambda = \frac{\beta_1^2 S_{xx}}{\sigma^2}$$

This noncentrality parameter also indicates that the observed value of F_0 should be large if $\beta_1 \neq 0$. Therefore, to test the hypothesis H_0: $\beta_1 = 0$, compute the test statistic F_0 and reject H_0 if

$$F_0 > F_{\alpha,1,n-2}$$

The test procedure is summarized in Table 2.4.

TABLE 2.4 Analysis of Variance for Testing Significance of Regression

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Regression	$SS_R = \hat{\beta}_1 S_{xy}$	1	MS_R	MS_R/MS_{Res}
Residual	$SS_{Res} = SS_T - \hat{\beta}_1 S_{xy}$	$n-2$	MS_{Res}	
Total	SS_T	$n-1$		

Example 2.4 The Rocket Propellant Data

We will test for significance of regression in the model developed in Example 2.1 for the rocket propellant data. The fitted model is $\hat{y} = 2627.82 - 37.15x$, $SS_T = 1{,}693{,}737.60$, and $S_{xy} = -41{,}112.65$. The regression sum of squares is computed from Eq. (2.34) as

$$SS_R = \hat{\beta}_1 S_{xy} = (-37.15)(-41{,}112.65) = 1{,}527{,}334.95$$

The analysis of variance is summarized in Table 2.5. The computed value of F_0 is 165.21, and from Table A.4, $F_{0.01,1,18} = 8.29$. The P value for this test is 1.66×10^{-10}. Consequently, we reject H_0: $\beta_1 = 0$. ■

Minitab Output The Minitab output in Table 2.3 also presents the analysis-of-variance test significance of regression. Comparing Tables 2.3 and 2.5, we note that there are some slight differences between the manual calculations and those performed by computer for the sums of squares. This is due to rounding the manual calculations to two decimal places. The computed values of the test statistics essentially agree.

More About the t Test We noted in Section 2.3.2 that the t statistic

$$t_0 = \frac{\hat{\beta}_1}{\mathrm{se}(\hat{\beta}_1)} = \frac{\hat{\beta}_1}{\sqrt{MS_{\mathrm{Res}}/S_{xx}}} \tag{2.37}$$

could be used for testing for significance of regression. However, note that on squaring both sides of Eq. (2.37), we obtain

$$t_0^2 = \frac{\hat{\beta}_1^2 S_{xx}}{MS_{\mathrm{Res}}} = \frac{\hat{\beta}_1 S_{xy}}{MS_{\mathrm{Res}}} = \frac{MS_R}{MS_{\mathrm{Res}}} \tag{2.38}$$

Thus, t_0^2 in Eq. (2.38) is identical to F_0 of the analysis-of-variance approach in Eq. (2.36). For example; in the rocket propellant example $t_0 = -12.5$, so $t_0^2 = (-12.5)^2 = 165.12 \simeq F_0 = 165.21$. In general, the square of a t random variable with f degrees of freedom is an F random variable with one and f degrees of freedom in the numerator and denominator, respectively. Although the t test for H_0: $\beta_1 = 0$ is equivalent to the F test in simple linear regression, the t test is somewhat more adaptable, as it could be used for one-sided alternative hypotheses (either H_1: $\beta_1 < 0$ or H_1: $\beta_1 > 0$), while the F test considers only the two-sided alternative. Regression

TABLE 2.5 Analysis-of-Variance Table for the Rocket Propellant Regression Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P value
Regression	1,527,334.95	1	1,527,334.95	165.21	1.66×10^{-10}
Residual	166,402.65	18	9,244.59		
Total	1,693,737.60	19			

computer programs routinely produce both the analysis of variance in Table 2.4 and the t statistic. Refer to the Minitab output in Table 2.3.

The real usefulness of the analysis of variance is in **multiple regression models**. We discuss multiple regression in the next chapter.

Finally, remember that deciding that $\beta_1 = 0$ is a very important conclusion that is only **aided** by the t or F test. The inability to show that the slope is not statistically different from zero may not necessarily mean that y and x are unrelated. It may mean that our ability to detect this relationship has been obscured by the variance of the measurement process or that the range of values of x is inappropriate. A great deal of nonstatistical evidence and knowledge of the subject matter in the field is required to conclude that $\beta_1 = 0$.

2.4 INTERVAL ESTIMATION IN SIMPLE LINEAR REGRESSION

In this section we consider confidence interval estimation of the regression model parameters. We also discuss interval estimation of the mean response $E(y)$ for given values of x. The normality assumptions introduced in Section 2.3 continue to apply.

2.4.1 Confidence Intervals on β_0, β_1, and σ^2

In addition to point estimates of β_0, β_1, and σ^2, we may also obtain confidence interval estimates of these parameters. The width of these confidence intervals is a measure of the overall quality of the regression line. If the errors are normally and independently distributed, then the sampling distribution of both $\left(\hat{\beta}_1 - \beta_1\right)/\mathrm{se}\left(\hat{\beta}_1\right)$ and $\left(\hat{\beta}_0 - \beta_0\right)/\mathrm{se}\left(\hat{\beta}_0\right)$ is t with $n - 2$ degrees of freedom. Therefore, a $100(1 - \alpha)$ percent confidence interval (CI) on the slope β_1 is given by

$$\hat{\beta}_1 - t_{\alpha/2,n-2}\mathrm{se}\left(\hat{\beta}_1\right) \le \hat{\beta}_1 \le \hat{\beta}_1 + t_{\alpha/2,n-2}\mathrm{se}\left(\hat{\beta}_1\right) \tag{2.39}$$

and a $100(1 - \alpha)$ percent CI on the intercept β_0 is

$$\hat{\beta}_0 - t_{\alpha/2,n-2}\mathrm{se}\left(\hat{\beta}_0\right) \le \beta_0 \le \hat{\beta}_0 + t_{\alpha/2,n-2}\mathrm{se}\left(\hat{\beta}_0\right) \tag{2.40}$$

These CIs have the usual frequentist interpretation. That is, if we were to take repeated samples of the same size at the same x levels and construct, for example, 95% CIs on the slope for each sample, then 95% of those intervals will contain the true value of β_1.

If the errors are normally and independently distributed, Appendix C.3 shows that the sampling distribution of $(n - 2)MS_{\mathrm{Res}}/\sigma^2$ is chi square with $n - 2$ degrees of freedom. Thus,

$$P\left\{\chi^2_{1-\alpha/2,n-2} \le \frac{(n-2)MS_{\mathrm{Res}}}{\sigma^2} \le \chi^2_{\alpha/2,n-2}\right\} = 1 - \alpha$$

and consequently a $100(1 - \alpha)$ percent CI on σ^2 is

$$\frac{(n-2)MS_{\text{Res}}}{\chi^2_{\alpha/2,n-2}} \le \sigma^2 \le \frac{(n-2)MS_{\text{Res}}}{\chi^2_{1-\alpha/2,n-2}} \tag{2.41}$$

Example 2.5 The Rocket Propellant Data

We construct 95% CIs on β_1 and σ^2 using the rocket propellant data from Example 2.1. The standard error of $\hat{\beta}_1$ is $\text{se}(\hat{\beta}_1) = 2.89$ and $t_{0.025,18} = 2.101$. Therefore, from Eq. (2.35), the 95% CI on the slope is

$$\hat{\beta}_1 - t_{0.025,18}\text{se}(\hat{\beta}_1) \le \beta_1 \le \hat{\beta}_1 + t_{0.025,18}\text{se}(\hat{\beta}_1)$$
$$-37.15 - (2.101)(2.89) \le \beta_1 \le -37.15 + (2.101)(2.89)$$

or

$$-43.22 \le \beta_1 \le -31.08$$

In other words, 95% of such intervals will include the true value of the slope.

If we had chosen a different value for α, the width of the resulting CI would have been different. For example, the 90% CI on β_1 is $-42.16 \le \beta_1 \le -32.14$, which is narrower than the 95% CI. The 99% CI is $-45.49 \le \beta_1 \le 28.81$, which is wider than the 95% CI. In general, the larger the confidence coefficient $(1 - \alpha)$ is, the wider the CI.

The 95% CI on σ^2 is found from Eq. (2.41) as follows:

$$\frac{(n-2)MS_{\text{Res}}}{\chi^2_{0.025,n-2}} \le \sigma^2 \le \frac{(n-2)MS_{\text{Res}}}{\chi^2_{0.975,n-2}}$$
$$\frac{18(9244.59)}{\chi^2_{0.025,18}} \le \sigma^2 \le \frac{18(9244.59)}{\chi^2_{0.975,18}}$$

From Table A.2, $\chi^2_{0.025,18} = 31.5$ and $\chi^2_{0.975,18} = 8.23$. Therefore, the desired CI becomes

$$\frac{18(9244.59)}{31.5} \le \sigma^2 \le \frac{18(9244.59)}{8.23}$$

or

$$5282.62 \le \sigma^2 \le 20{,}219.03$$ ■

2.4.2 Interval Estimation of the Mean Response

A major use of a regression model is to estimate the mean response $E(y)$ for a particular value of the regressor variable x. For example, we might wish to estimate the mean shear strength of the propellant bond in a rocket motor made from a batch of sustainer propellant that is 10 weeks old. Let x_0 be the level of the regressor variable for which we wish to estimate the mean response, say $E(y|x_0)$. We assume that

x_0 is any value of the regressor variable within the range of the original data on x used to fit the model. An unbiased point estimator of $E(y|x_0)$ is found from the fitted model as

$$\widehat{E(y|x_0)} = \hat{\mu}_{y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0 \tag{2.42}$$

To obtain a 100(1 − α) percent CI on $E(y|x_0)$, first note that $\hat{\mu}_{y|x_0}$ is a normally distributed random variable because it is a linear combination of the observations y_i. The variance of $\hat{\mu}_{y|x_0}$ is

$$\begin{aligned}\operatorname{Var}(\hat{\mu}_{y|x_0}) &= \operatorname{Var}\left(\hat{\beta}_0 + \hat{\beta}_1 x_0\right) = \operatorname{Var}\left[\bar{y} + \hat{\beta}_1(x_0 - \bar{x})\right] \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2 (x_0 - \bar{x})^2}{S_{xx}} = \sigma^2\left[\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]\end{aligned}$$

since (as noted in Section 2.2.4) $\operatorname{Cov}\left(\bar{y}, \hat{\beta}_1\right) = 0$. Thus, the sampling distribution of

$$\frac{\hat{\mu}_{y|x_0} - E(y|x_0)}{\sqrt{MS_{\text{Res}}\left(1/n + (x_0 - \bar{x})^2/S_{xx}\right)}}$$

is t with $n - 2$ degrees of freedom. Consequently, **a 100(1 − α) percent CI on the mean response at the point $x = x_0$** is

$$\begin{aligned}&\hat{\mu}_{y|x_0} - t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)} \\ &\quad \le E(y|x_0) \le \hat{\mu}_{y|x_0} + t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}\end{aligned} \tag{2.43}$$

Note that the width of the CI for $E(y|x_0)$ is a function of x_0. The interval width is a minimum for $x_0 = \bar{x}$ and widens as $|x_0 - \bar{x}|$ increases. Intuitively this is reasonable, as we would expect our best estimates of y to be made at x values near the center of the data and the precision of estimation to deteriorate as we move to the boundary of the x space.

Example 2.6 The Rocket Propellant Data

Consider finding a 95% CI on $E(y|x_0)$ for the rocket propellant data in Example 2.1. The CI is found from Eq. (2.43) as

$$\begin{aligned}&\hat{\mu}_{y|x_0} - t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)} \\ &\quad \le E(y|x_0) \le \hat{\mu}_{y|x_0} + t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}\end{aligned}$$

$$\hat{\mu}_{y|x_0} - (2.101)\sqrt{9244.59\left(\frac{1}{20} + \frac{(x_0 - 13.3625)^2}{1106.56}\right)}$$
$$\le E(y|x_0) \le \hat{\mu}_{y|x_0} + (2.101)\sqrt{9244.59\left(\frac{1}{20} + \frac{(x_0 - 13.3625)^2}{1106.56}\right)}$$

If we substitute values of x_0 and the fitted value $\hat{y}_0 = \hat{\mu}_{y|x_0}$ at the value of x_0 into this last equation, we will obtain the 95% CI on the mean response at $x = x_0$. For example, if $x_0 = \bar{x} = 13.3625$, then $\hat{\mu}_{y|x_0} = 2131.40$, and the CI becomes

$$2086.230 \le E(y|13.3625) \le 2176.571$$

Table 2.6 contains the 95% confidence limits on $E(y|x_0)$ for several other values of x_0. These confidence limits are illustrated graphically in Figure 2.4. Note that the width of the CI increases as $|x_0 - \bar{x}|$ increases. ■

TABLE 2.6 Confidence Limits on $E(y|x_0)$ for Several Values of x_0

Lower Confidence Limit	x_0	Upper Confidence Limit
2438.919	3	2593.821
2341.360	6	2468.481
2241.104	9	2345.836
2136.098	12	2227.942
2086.230	$\bar{x} = 13.3625$	2176.571
2024.318	15	2116.822
1905.890	18	2012.351
1782.928	21	1912.412
1657.395	24	1815.045

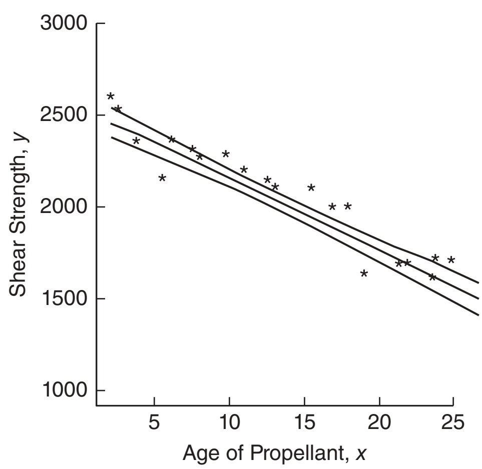


Figure 2.4 The upper and lower 95% confidence limits for the propellant data.

Many regression textbooks state that one should never use a regression model to **extrapolate** beyond the range of the original data. By extrapolation, we mean using the prediction equation beyond the boundary of the x space. Figure 1.5 illustrates clearly the dangers inherent in extrapolation; model or equation error can severely damage the prediction.

Equation (2.43) points out that the issue of extrapolation is much more subtle; the further the x value is from the center of the data, the more variable our estimate of $E(y|x_0)$. Please note, however, that nothing "magical" occurs at the boundary of the x space. It is not reasonable to think that the prediction is wonderful at the observed data value most remote from the center of the data and completely awful just beyond it. Clearly, Eq. (2.43) points out that we should be concerned about prediction quality as we approach the boundary and that as we move beyond this boundary, the prediction may deteriorate rapidly. Furthermore, the farther we move away from the original region of x space, the more likely it is that equation or model error will play a role in the process.

This is not the same thing as saying "never extrapolate." Engineers and economists routinely use prediction equations to forecast a variable of interest one or more time periods in the future. Strictly speaking, this forecast is an extrapolation. Equation (2.43) supports such use of the prediction equation. However, Eq. (2.43) does not support using the regression model to forecast many periods in the future. Generally, the greater the extrapolation, the higher is the chance of equation error or model error impacting the results.

The probability statement associated with the CI (2.43) holds only when a single CI on the mean response is to be constructed. A procedure for constructing several CIs that, considered jointly, have a specified confidence level is a **simultaneous statistical inference** problem. These problems are discussed in Chapter 3.

2.5 PREDICTION OF NEW OBSERVATIONS

An important application of the regression model is prediction of new observations y corresponding to a specified level of the regressor variable x. If x_0 is the value of the regressor variable of interest, then

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 \tag{2.44}$$

is the point estimate of the new value of the response y_0.

Now consider obtaining an interval estimate of this future observation y_0. The CI on the mean response at $x = x_0$ [Eq. (2.43)] is inappropriate for this problem because it is an interval estimate on the **mean** of y (a parameter), not a probability statement about future observations from that distribution. We now develop a **prediction interval for the future observation y_0**.

Note that the random variable

$$\psi = y_0 - \hat{y}_0$$

is normally distributed with mean zero and variance

$$\mathrm{Var}(\psi) = \mathrm{Var}(y_0 - \hat{y}_0) = \sigma^2\left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right]$$

because the future observation y_0 is independent of $\hat{y}_0$. If we use $\hat{y}_0$ to predict y_0, then the standard error of $\psi = y_0 - \hat{y}_0$ is the appropriate statistic on which to base a prediction interval. Thus, the $100(1 - \alpha)$ percent prediction interval on a future observation at x_0 is

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{MS_{\mathrm{Res}}\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)} \\ \leq y_0 \leq \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{MS_{\mathrm{Res}}\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)} \tag{2.45}$$

The prediction interval (2.45) is of minimum width at $x_0 = \bar{x}$ and widens as $|x_0 - \bar{x}|$ increases. By comparing (2.45) with (2.43), we observe that the prediction interval at x_0 is always wider than the CI at x_0 because the prediction interval depends on both the error from the fitted model and the error associated with future observations.

Example 2.7 The Rocket Propellant Data

We find a 95% prediction interval on a future value of propellant shear strength in a motor made from a batch of sustainer propellant that is 10 weeks old. Using (2.45), we find that the prediction interval is

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{MS_{\mathrm{Res}}\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)} \\ \leq y_0 \leq \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{MS_{\mathrm{Res}}\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}$$

$$2256.32 - (2.101)\sqrt{9244.59\left(1 + \frac{1}{20} + \frac{(10 - 13.3625)^2}{1106.56}\right)} \\ \leq y_0 \leq 2256.32 + (2.101)\sqrt{9244.59\left(1 + \frac{1}{20} + \frac{(10 - 13.3625)^2}{1106.56}\right)}$$

which simplifies to

$$2048.32 \leq y_0 \leq 2464.32$$

Therefore, a new motor made from a batch of 10-week-old sustainer propellant could reasonably be expected to have a propellant shear strength between 2048.32 and 2464.32 psi.

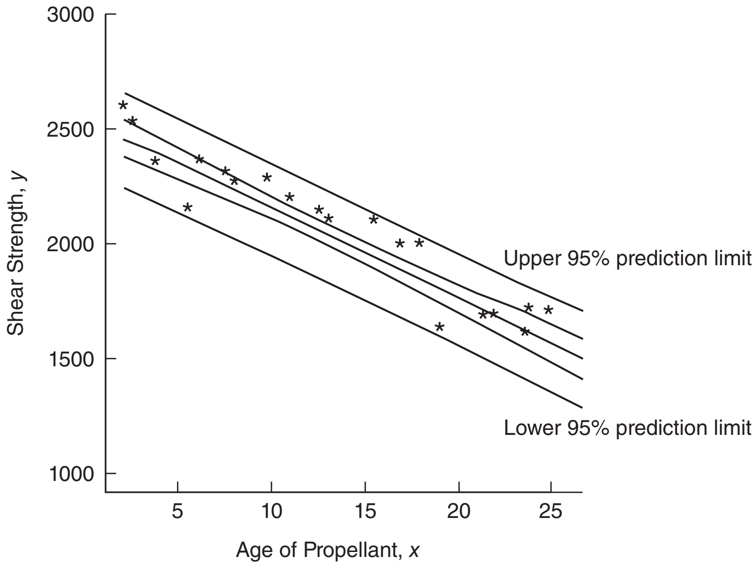


Figure 2.5 The 95% confidence and prediction intervals for the propellant data.

Figure 2.5 shows the 95% prediction interval calculated from (2.45) for the rocket propellant regression model. Also shown on this graph is the 95% CI on the mean [that is, $E(y|x)$ from Eq. (2.43). This graph nicely illustrates the point that the prediction interval is wider than the corresponding CI. ■

We may generalize (2.45) somewhat to find a 100(1 − α) percent prediction interval on the **mean** of m future observations on the response at $x = x_0$. Let $\bar{y}_0$ be the mean of m future observations at $x = x_0$. A point estimator of $\bar{y}_0$ is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. The 100(1 − α)% prediction interval on $\bar{y}_0$ is

$$\hat{y}_0 - t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{m}+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$
$$\le y_0 \le \hat{y}_0 + t_{\alpha/2,n-2}\sqrt{MS_{\text{Res}}\left(\frac{1}{m}+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)} \tag{2.46}$$

2.6 COEFFICIENT OF DETERMINATION

The quantity

$$R^2 = \frac{SS_{\text{R}}}{SS_{\text{T}}} = 1 - \frac{SS_{\text{Res}}}{SS_{\text{T}}} \tag{2.47}$$

is called the **coefficient of determination**. Since SS_{T} is a measure of the variability in y without considering the effect of the regressor variable x and SS_{Res} is a measure of the variability in y remaining after x has been considered, R^2 is often called the proportion of variation explained by the regressor x. Because $0 \le SS_{\mathrm{Res}} \le SS_{\mathrm{T}}$, it follows that $0 \le R^2 \le 1$. Values of R^2 that are close to 1 imply that most of the variability in y is explained by the regression model. For the regression model for the rocket propellant data in Example 2.1, we have

$$R^2 = \frac{SS_{\mathrm{R}}}{SS_{\mathrm{T}}} = \frac{1,527,334.95}{1,693,737.60} = 0.9018$$

that is, 90.18% of the variability in strength is accounted for by the regression model.

The statistic R^2 should be used with caution, since it is always possible to make R^2 large by adding enough terms to the model. For example, if there are no repeat points (more than one y value at the same x value), a polynomial of degree $n - 1$ will give a "perfect" fit ($R^2 = 1$) to n data points. When there are repeat points, R^2 can never be exactly equal to 1 because the model cannot explain the variability related to "pure" error.

Although R^2 cannot decrease if we add a regressor variable to the model, this does not necessarily mean the new model is superior to the old one. Unless the error sum of squares in the new model is reduced by an amount equal to the original error mean square, the new model will have a larger error mean square than the old one because of the loss of one degree of freedom for error. Thus, the new model will actually be worse than the old one.

The magnitude of R^2 also depends on the range of variability in the regressor variable. Generally R^2 will increase as the spread of the x's increases and decrease as the spread of the x's decreases provided the assumed model form is correct. By the delta method (also see Hahn 1973), one can show that the expected value of R^2 from a straight-line regression is approximately

$$E\left(R^2\right) \approx \frac{\beta_1^2 S_{xx}/n-1}{\dfrac{\beta_1^2 S_{xx}}{n-1} + \sigma^2}$$

Clearly the expected value of R^2 will increase (decrease) as S_{xx} (a measure of the spread of the x's) increases (decreases). Thus, a large value of R^2 may result simply because x has been varied over an unrealistically large range. On the other hand, R^2 may be small because the range of x was too small to allow its relationship with y to be detected.

There are several other misconceptions about R^2. In general, R^2 does not measure the magnitude of the slope of the regression line. A large value of R^2 does not imply a steep slope. Furthermore, R^2 does not measure the appropriateness of the linear model, for R^2 will often be large even though y and x are nonlinearly related. For example, R^2 for the regression equation in Figure 2.3*b* will be relatively large even though the linear approximation is poor. Remember that although R^2 is large, this does not necessarily imply that the regression model will be an accurate predictor.

2.7 A SERVICE INDUSTRY APPLICATION OF REGRESSION

A hospital is implementing a program to improve service quality and productivity. As part of this program the hospital management is attempting to measure and evaluate patient satisfaction. Table B.17 contains some of the data that have been collected on a random sample of 25 recently discharged patients. The response variable is satisfaction, a subjective response measure on an increasing scale. The potential regressor variables are patient age, severity (an index measuring the severity of the patient's illness), an indicator of whether the patient is a surgical or medical patient (0 = surgical, 1 = medical), and an index measuring the patient's anxiety level. We start by building a simple linear regression model relating the response variable satisfaction to severity.

Figure 2.6 is a scatter diagram of satisfaction versus severity. There is a relatively mild indication of a potential linear relationship between these two variables. The output from JMP for fitting a simple linear regression model to these data is shown in Figure 2.7. JMP is an SAS product that is a menu-based PC statistics package with an extensive array of regression modeling and analysis capabilities.

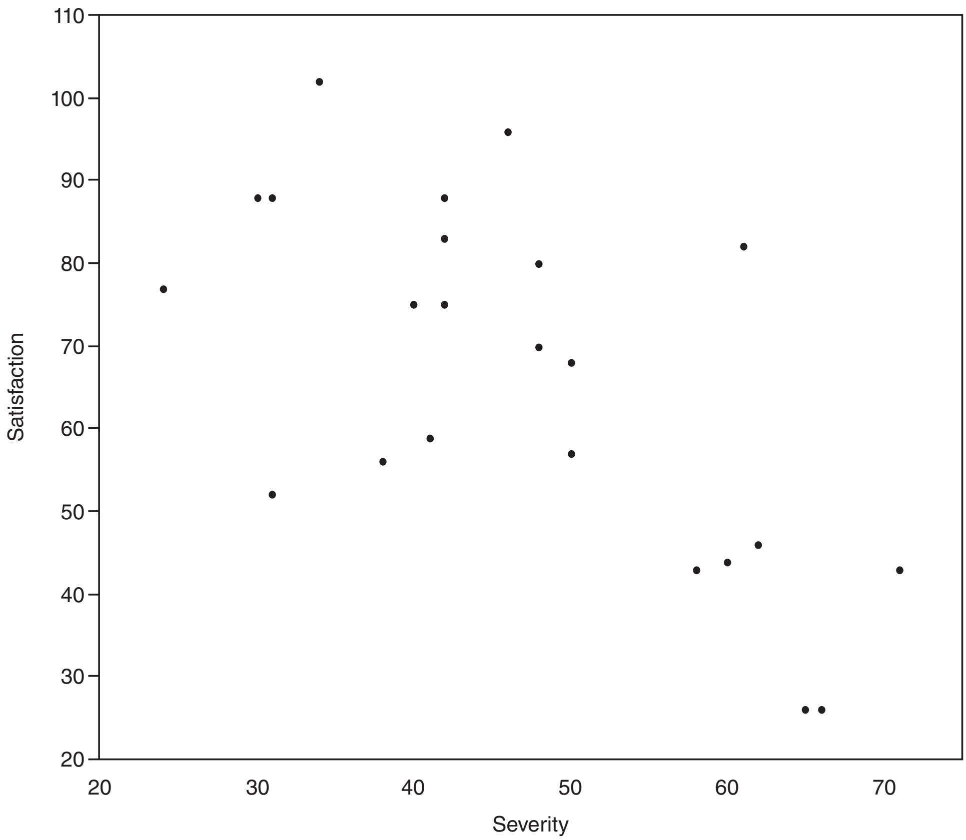
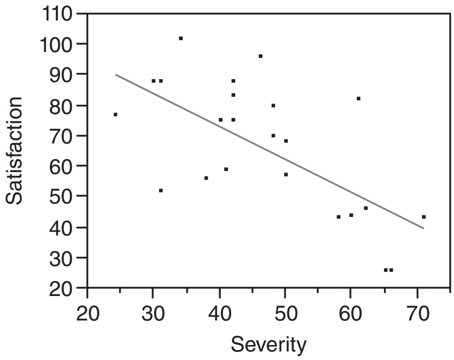
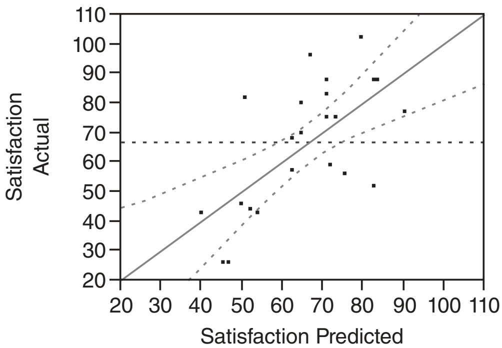


Figure 2.6 Scatter diagram of satisfaction versus severity.

Response Satisfaction
Whole Model
Regression Plot



Actual by Predicted Plot



Summary of Fit

RSquare	0.426596
RSquare Adj	0.401666
Root Mean Square Error	16.43242
Mean of Response	66.72
Observations (or Sum Wgts)	25

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	4620.482	4620.48	17.1114
Error	23	6210.558	270.02	**Prob > F**
C. Total	24	10831.040		0.0004*

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	115.6239	12.27059	9.42	<.0001*

Figure 2.7 JMP output for the simple linear regression model for the patient satisfaction data.

At the top of the JMP output is the scatter plot of the satisfaction and severity data, along with the fitted regression line. The straight line fit looks reasonable although there is considerable variability in the observations around the regression line. The second plot is a graph of the actual satisfaction response versus the predicted response. If the model were a perfect fit to the data all of the points in this plot would lie exactly along the 45-degree line. Clearly, this model does not provide a perfect fit. Also, notice that while the regressor variable is significant (the ANOVA F statistic is 17.1114 with a P value that is less than 0.0004), the coefficient of determination $R^2 = 0.43$. That is, the model only accounts for about 43% of the variability in the data. It can be shown by the methods discussed in Chapter 4 that there are no fundamental problems with the underlying assumptions or measures of model adequacy, other than the rather low value of R^2.

Low values for R^2 occur occasionally in practice. The model is significant, there are no obvious problems with assumptions or other indications of model inadequacy, but the proportion of variability explained by the model is low. Now this is not an entirely disastrous situation. There are many situations where explaining 30 to 40% of the variability in y with a single predictor provides information of considerable value to the analyst. Sometimes, a low value of R^2 results from having a lot of variability in the measurements of the response due to perhaps the type of measuring instrument being used, or the skill of the person making the measurements. Here the variability in the response probably arises because the response is an expression of opinion, which can be very subjective. Also, the measurements are taken on human patients, and there can be considerably variability both within people and between people. Sometimes, a low value of R^2 is a result of a poorly specified model. In these cases the model can often be improved by the addition of one or more predictor or regressor variables. We see in Chapter 3 that the addition of another regressor results in considerable improvement of this model.

2.8 USING SAS® AND R FOR SIMPLE LINEAR REGRESSION

The purpose of this section is to introduce readers to SAS and to R. Appendix D gives more details about using SAS, including how to import data from both text and EXCEL files. Appendix E introduces the R statistical software package. R is becoming increasingly popular since it is free over the Internet.

Table 2.7 gives the SAS source code to analyze the rocket propellant data that we have been analyzing throughout this chapter. Appendix D provides detail explaining how to enter the data into SAS. The statement PROC REG tells the software that we wish to perform an ordinary least-squares linear regression analysis. The "model" statement specifies the specific model and tells the software which analyses to perform. The variable name to the left of the equal sign is the response. The variables to the right of the equal sign but before the solidus are the regressors. The information after the solidus specifies additional analyses. By default, SAS prints the analysis-of-variance table and the tests on the individual coefficients. In this case, we have specified three options: "p" asks SAS to print the predicted values, "clm" (which stands for confidence limit, mean) asks SAS to print the confidence band, and "cli" (which stands for confidence limit, individual observations) asks SAS to print the prediction band.

TABLE 2.7 SAS Code for Rocket Propellant Data

```
data rocket;
input shear age;
cards;
2158.70 15.50
1678.15 23.75
2316.00  8.00
2061.30 17.00
2207.50  5.50
1708.30 19.00
1784.70 24.00
2575.00  2.50
2357.90  7.50
2256.70 11.00
2165.20 13.00
2399.55  3.75
1779.80 25.99
2336.75  9.75
1765.30 22.00
2053.50 18.00
2414.40  6.00
2200.50 12.50
2654.20  2.00
1753.70 21.50
proc reg;
model shear=age/p clm cli;
run;
```

Table 2.8 gives the SAS output for this analysis. PROC REG always produces the analysis-of-variance table and the information on the parameter estimates. The “p clm cli” options on the model statement produced the remainder of the output file.

SAS also produces a log file that provides a brief summary of the SAS session. The log file is almost essential for debugging SAS code. Appendix D provides more details about this file.

R is a popular statistical software package, primarily because it is freely available at www.r-project.org. An easier-to-use version of R is R Commander. R itself is a high-level programming language. Most of its commands are prewritten functions. It does have the ability to run loops and call other routines, for example, in C. Since it is primarily a programming language, it often presents challenges to novice users. The purpose of this section is to introduce the reader as to how to use R to analyze simple linear regression data sets.

The first step is to create the data set. The easiest way is to input the data into a text file using spaces for delimiters. Each row of the data file is a record. The top row should give the names for each variable. All other rows are the actual data records. For example, consider the rocket propellant data from Example 2.1 given in Table 2.1. Let propellant.txt be the name of the data file. The first row of the text file gives the variable names:

TABLE 2.8 SAS Output for Analysis of Rocket Propellant Data.

SAS system 1

The REG Procedure
Model: MODEL1
Dependent Variable: shear

Number of Observations Read	20
Number of Observations Used	20

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1527483	1527483	165.38	<.0001
Error	18	166255	9236.38100		
Corrected Total	19	1693738			

Root MSE	96.10609	R- square	0.9018
Dependent Mean	2131.35750	Adj R- Sq	0.8964
Coeff Var	4.50915		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t value	Pr > |t|
Intercept	1	2627.82236	44.18391	59.47	<.0001
age	1	−37.15359	2.88911	−12.86	<.0001

The SAS System 2

The REG Procedure
Model: MODEL1
Dependent Variable: shear

Output Statistics

Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	95% CL Mean		95% CL Predict		Residual
1	2159	2052	22.3597	2005	2099	1845	2259	106.7583
2	1678	1745	36.9114	1668	1823	1529	1962	−67.2746
3	2316	2331	26.4924	2275	2386	2121	2540	−14.5936
4	2061	1996	23.9220	1946	2046	1788	2204	65.0887
5	2208	2423	31.2701	2358	2489	2211	2636	−215.9776
6	1708	1922	26.9647	1865	1979	1712	2132	−213.6041
7	1785	1736	37.5010	1657	1815	1519	1953	48.5638
8	2575	2535	38.0356	2455	2615	2318	2752	40.0616
9	2358	2349	27.3623	2292	2407	2139	2559	8.7296
10	2257	2219	22.5479	2172	2267	2012	2427	37.5671
11	2165	2145	21.5155	2100	2190	1938	2352	20.3743
12	2400	2488	35.1152	2415	2562	2274	2703	−88.9464
13	1780	1699	39.9031	1615	1783	1480	1918	80.8174
14	2337	2266	23.8903	2215	2316	2058	2474	71.1752
15	1765	1810	32.9362	1741	1880	1597	2024	−45.1434
16	2054	1959	25.3245	1906	2012	1750	2168	94.4423
17	2414	2405	30.2370	2341	2468	2193	2617	9.4992
18	2201	2163	21.6340	2118	2209	1956	2370	37.0975
19	2654	2554	39.2360	2471	2636	2335	2772	100.6848
20	1754	1829	31.8519	1762	1896	1616	2042	−75.3202

Sum of Residuals	0
Sum of squared Residuals	166255
Predicted Residual SS (PRESS)	205944

```
strength age
```

The next row is the first data record, with spaces delimiting each data item:

```
2158.70 15.50
```

The R code to read the data into the package is:

```
prop <- read.table("propellant.txt",header=TRUE, sep="")
```

The object prop is the R data set, and "propellant.txt" is the original data file. The phrase, header=TRUE tells R that the first row is the variable names. The phrase sep="" tells R that the data are space delimited.

The commands

```
prop.model <- lm(strength~age, data=prop)
summary(prop.model)
```

tell R

- to estimate the model, and
- to print the analysis of variance, the estimated coefficients, and their tests.

R Commander is an add-on package to R. It also is freely available. It provides an easy-to-use user interface, much like Minitab and JMP, to the parent R product. R Commander makes it much more convenient to use R; however, it does not provide much flexibility in its analysis. R Commander is a good way for users to get familiar with R. Ultimately, however, we recommend the use of the parent R product.

2.9 SOME CONSIDERATIONS IN THE USE OF REGRESSION

Regression analysis is widely used and, unfortunately, frequently misused. There are several common abuses of regression that should be mentioned:

1. Regression models are intended as interpolation equations over the range of the regressor variable(s) used to fit the model. As observed previously, we must be careful if we extrapolate outside of this range. Refer to Figure 1.5.
2. The disposition of the x values plays an important role in the least-squares fit. While all points have equal weight in determining the height of the line, the slope is more strongly influenced by the remote values of x. For example, consider the data in Figure 2.8. The slope in the least-squares fit depends heavily on either or both of the points A and B. Furthermore, the remaining data would give a very different estimate of the slope if A and B were deleted. Situations

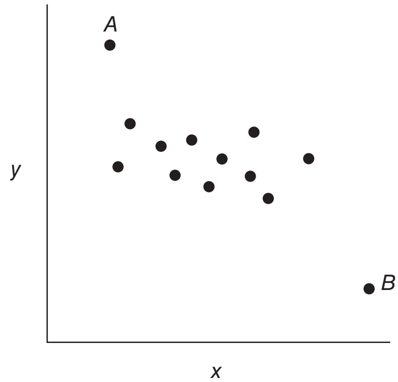


Figure 2.8 Two influential observations.

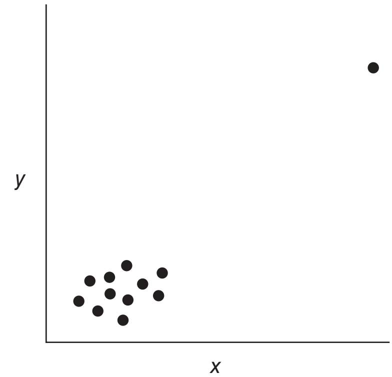


Figure 2.9 A point remote in x space.

such as this often require corrective action, such as further analysis and possible deletion of the unusual points, estimation of the model parameters with some technique that is less seriously influenced by these points than least squares, or restructuring the model, possibly by introducing further regressors.

A somewhat different situation is illustrated in Figure 2.9, wher one of the 12 observations is very remote in x space. In this example the slope is largely determined by the extreme point. If this point is deleted, the slope estimate is probably zero. Because of the gap between the two clusters of points, we really have only two distinct information units with which to fit the model. Thus, there are effectively far fewer than the apparent 10 degrees of freedom for error.

Situations such as these seem to occur fairly often in practice. In general we should be aware that in some data sets one point (or a small cluster of points) may control key model properties.

3. **Outliers** are observations that differ considerably from the rest of the data. They can seriously disturb the least-squares fit. For example, consider the data in Figure 2.10. Observation A seems to be an outlier because it falls far from the line implied by the rest of the data. If this point is really an outlier, then the estimate of the intercept may be incorrect and the residual mean square may be an inflated estimate of σ^2. The outlier may be a "bad value" that has resulted from a data recording or some other error. On the other hand, the data point may not be a bad value and may be a highly useful piece of evidence concerning the process under investigation. Methods for detecting and dealing with outliers are discussed more completely in Chapter 4.
4. As mentioned in Chapter 1, just because a regression analysis has indicated a strong relationship between two variables, this does not imply that the variables are related in any causal sense. Causality implies necessary correlation. Regression analysis can only address the issues on correlation. It cannot address the issue of necessity. Thus, our expectations of discovering cause-and-effect relationships from regression should be modest.

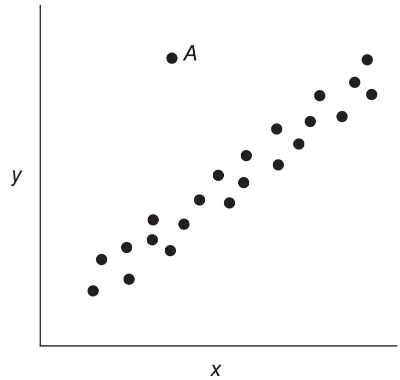


Figure 2.10 An outlier.

TABLE 2.9 Data Illustrating Nonsense Relationships between Variables

Year	Number of Certified Mental Defectives per 10,000 of Estimated Population in the U.K (y)	Number of Radio Receiver Licenses Issued (Millions) in the U.K (x_1)	First Name of President of the U.S. (x_2)
1924	8	1.350	Calvin
1925	8	1.960	Calvin
1926	9	2.270	Calvin
1927	10	2.483	Calvin
1928	11	2.730	Calvin
1929	11	3.091	Calvin
1930	12	3.647	Herbert
1931	16	4.620	Herbert
1932	18	5.497	Herbert
1933	19	6.260	Herbert
1934	20	7.012	Franklin
1935	21	7.618	Franklin
1936	22	8.131	Franklin
1937	23	8.593	Franklin

Source: Kendall and Yule [1950] and Tufte [1974].

As an example of a "nonsense" relationship between two variables, consider the data in Table 2.9. This table presents the number of certified mental defectives in the United Kingdom per 10,000 of estimated population (y), the number of radio receiver licenses issued (x_1), and the first name of the President of the United States (x_2) for the years 1924–1937. We can show that the regression equation relating y to x_1 is

$$\hat{y} = 4.582 + 2.204x_1$$

The t statistic for testing H_0: $\beta_1 = 0$ for this model is $t_0 =$ 27.312 (the P value is 3.58×10^{-12}), and the coefficient of determination is $R^2 = 0.9842$. That is, 98.42% of the variability in the data is explained by the number of radio receiver licenses issued. Clearly this is a nonsense relationship, as it is highly unlikely that the number of mental defectives in the population is functionally related to the number of radio receiver licenses issued. The reason for this strong statistical relationship is that y and x_1 are monotonically related (two sequences of numbers are monotonically related if as one sequence increases, the other always either increases or decreases). In this example y is increasing because diagnostic procedures for mental disorders are becoming more refined over the years represented in the study and x_1 is increasing because of the emergence and low-cost availability of radio technology over the years.

Any two sequences of numbers that are monotonically related will exhibit similar properties. To illustrate this further, suppose we regress y on the number of letters in the first name of the U.S. president in the corresponding year. The model is

$$\hat{y} = -26.442 + 5.900x_2$$

with $t_0 = 8.996$ (the P value is 1.11×10^{-6}) and $R^2 = 0.8709$. Clearly this is a nonsense relationship as well.

5. In some applications of regression the value of the regressor variable x required to predict y is unknown. For example, consider predicting maximum daily load on an electric power generation system from a regression model relating the load to the maximum daily temperature. To predict tomorrow's maximum load, we must first predict tomorrow's maximum temperature. Consequently, the prediction of maximum load is **conditional** on the temperature forecast. The accuracy of the maximum load forecast depends on the accuracy of the temperature forecast. This must be considered when evaluating model performance.

Other abuses of regression are discussed in subsequent chapters. For further reading on this subject, see the article by Box [1966].

2.10 REGRESSION THROUGH THE ORIGIN

Some regression situations seem to imply that a straight line passing through the origin should be fit to the data. A **no-intercept regression model** often seems appropriate in analyzing data from chemical and other manufacturing processes. For example, the yield of a chemical process is zero when the process operating temperature is zero.

The no-intercept model is

$$y = \beta_1 x + \varepsilon \tag{2.48}$$

Given n observations (y_i, x_i), $i = 1, 2, \ldots, n$, the least-squares function is

$$S(\beta_1) = \sum_{i=1}^{n} (y_i - \beta_1 x_i)^2$$

The only normal equation is

$$\hat{\beta}_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} y_i x_i \tag{2.49}$$

and the **least-squares estimator of the slope** is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2} \tag{2.50}$$

The estimator of $\hat{\beta}_1$ is unbiased for β_1, and the **fitted regression model** is

$$\hat{y} = \hat{\beta}_1 x \tag{2.51}$$

The estimator of σ^2 is

$$\hat{\sigma}^2 = MS_{\text{Res}} = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n-1} = \frac{\sum_{i=1}^{n} y_i^2 - \hat{\beta}_1 \sum_{i=1}^{n} y_i x_i}{n-1} \tag{2.52}$$

with $n - 1$ degrees of freedom.

Making the normality assumption on the errors, we may test hypotheses and construct confidence and prediction intervals for the no-intercept model. The **100(1 – α) percent CI on β_1** is

$$\hat{\beta}_1 - t_{\alpha/2,n-1} \sqrt{\frac{MS_{\text{Res}}}{\sum_{i=1}^{n} x_i^2}} \le \beta_1 \le \hat{\beta}_1 + t_{\alpha/2,n-1} \sqrt{\frac{MS_{\text{Res}}}{\sum_{i=1}^{n} x_i^2}} \tag{2.53}$$

A **100(1 – α) percent CI on $E(y|x_0)$**, the mean response at $x = x_0$, is

$$\hat{\mu}_{y|x_0} - t_{\alpha/2,n-1} \sqrt{\frac{x_0^2 MS_{\text{Res}}}{\sum_{i=1}^{n} x_i^2}} \le E(y|x_0) \le \hat{\mu}_{y|x_0} + t_{\alpha/2,n-1} \sqrt{\frac{x_0^2 MS_{\text{Res}}}{\sum_{i=1}^{n} x_i^2}} \tag{2.54}$$

The **100(1 – α) percent prediction interval on a future observation at $x = x_0$**, say y_0, is

$$\hat{y}_0 - t_{\alpha/2,n-1}\sqrt{MS_{\mathrm{Res}}\left(1+\frac{x_0^2}{\sum_{i=1}^{n} x_i^2}\right)} \le y_0 \le \hat{y}_0 + t_{\alpha/2,n-1}\sqrt{MS_{\mathrm{Res}}\left(1+\frac{x_0^2}{\sum_{i=1}^{n} x_i^2}\right)} \tag{2.55}$$

Both the CI (2.54) and the prediction interval (2.55) widen as x_0 increases. Furthermore, the length of the CI (2.54) at $x = 0$ is zero because the model assumes that the mean y at $x = 0$ is known with certainty to be zero. This behavior is considerably different than observed in the intercept model. The prediction interval (2.55) has nonzero length at $x_0 = 0$ because the random error in the future observation must be taken into account.

It is relatively easy to misuse the no-intercept model, particularly in situations where the data lie in a region of x space remote from the origin. For example, consider the no-intercept fit in the scatter diagram of chemical process yield (y) and operating temperature (x) in Figure 2.11*a*. Although over the range of the regressor variable 100°F ≤ x ≤ 200°F, yield and temperature seem to be linearly related, forcing the model to go through the origin provides a visibly poor fit. A model containing an intercept, such as illustrated in Figure 2.11*b*, provides a much better fit in the region of x space where the data were collected.

Frequently the relationship between y and x is quite different near the origin than it is in the region of x space containing the data. This is illustrated in Figure 2.12 for the chemical process data. Here it would seem that either a quadratic or a more complex nonlinear regression model would be required to adequately express the relationship between y and x over the entire range of x. Such a model should only be entertained if the range of x in the data is sufficiently close to the origin.

The scatter diagram sometimes provides guidance in deciding whether or not to fit the no-intercept model. Alternatively we may fit both models and choose between them based on the quality of the fit. If the hypothesis $\beta_0 = 0$ cannot be rejected in the intercept model, this is an indication that the fit may be improved by using the

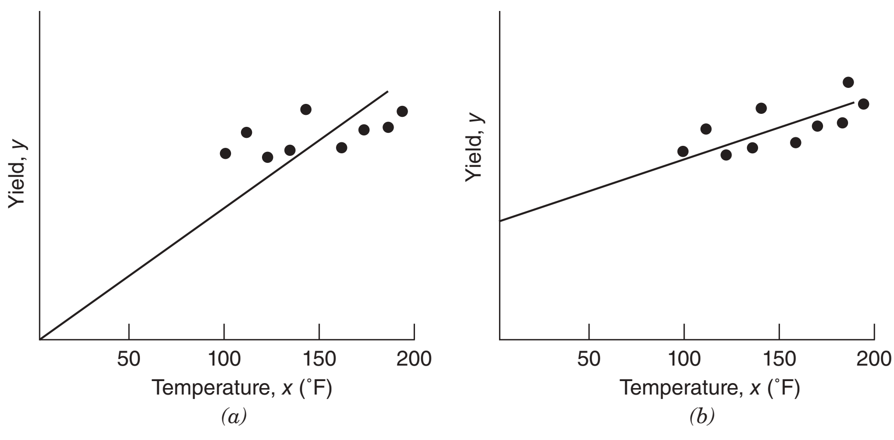


Figure 2.11 Scatter diagrams and regression lines for chemical process yield and operating temperature: (*a*) no-intercept model; (*b*) intercept model.

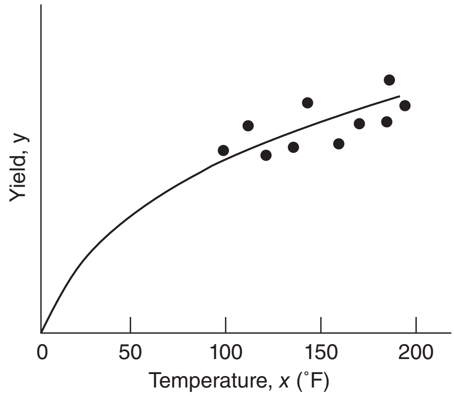


Figure 2.12 True relationship between yield and temperature.

no-intercept model. The residual mean square is a useful way to compare the quality of fit. The model having the smaller residual mean square is the best fit in the sense that it minimizes the estimate of the variance of y about the regression line.

Generally R^2 is not a good comparative statistic for the two models. For the intercept model we have

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = \frac{\text{variation in } y \text{ explained by regression}}{\text{total observed variation in } y}$$

Note that R^2 indicates the proportion of variability around $\bar{y}$ explained by regression. In the no-intercept case the fundamental analysis-of-variance identity (2.32) becomes

$$\sum_{i=1}^{n} y_i^2 = \sum_{i=1}^{n} \hat{y}_i^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

so that the no-intercept model analogue for R^2 would be

$$R_0^2 = \frac{\sum_{i=1}^{n} \hat{y}_i^2}{\sum_{i=1}^{n} y_i^2}$$

The statistic R_0^2 indicates the proportion of variability around the **origin** (zero) accounted for by regression. We occasionally find that R_0^2 is larger than R^2 even though the residual mean square (which is a reasonable measure of the overall quality of the fit) for the intercept model is smaller than the residual mean square for the no-intercept model. This arises because R_0^2 is computed using uncorrected sums of squares.

There are alternative ways to define R^2 for the no-intercept model. One possibility is

$$R_{0'}^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

However, in cases where $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ is large, $R_{0'}^2$ can be negative. We prefer to use MS_{Res} as a basis of comparison between intercept and no-intercept regression models. A nice article on regression models with no intercept term is Hahn [1979].

Example 2.8 The Shelf-Stocking Data

The time required for a merchandiser to stock a grocery store shelf with a soft drink product as well as the number of cases of product stocked is shown in Table 2.10. The scatter diagram shown in Figure 2.13 suggests that a straight line passing through the origin could be used to express the relationship between time and the number of cases stocked. Furthermore, since if the number of cases $x = 0$, then shelf stocking time $y = 0$, this model seems intuitively reasonable. Note also that the range of x is close to the origin.

The slope in the no-intercept model is computed from Eq. (2.50) as

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2} = \frac{1841.98}{4575.00} = 0.4026$$

Therefore, the fitted equation is

$$\hat{y} = 0.4026x$$

This regression line is shown in Figure 2.14. The residual mean square for this model is $MS_{\text{Res}} = 0.0893$ and $R_0^2 = 0.9883$. Furthermore, the t statistic for testing H_0: $\beta_1 = 0$ is $t_0 = 91.13$, for which the P value is 8.02×10^{-21}. These summary statistics do not reveal any startling inadequacy in the no-intercept model. ■

We may also fit the intercept model to the data for comparative purposes. This results in

$$\hat{y} = -0.0938 + 0.4071x$$

The t statistic for testing H_0: $\beta_0 = 0$ is $t_0 = -0.65$, which is not significant, implying that the no-intercept model may provide a superior fit. The residual mean square

TABLE 2.10 Shelf-Stocking Data for Example 2.8

Times, y (minutes)	Cases Stocked, x
10.15	25
2.96	6
3.00	8
6.88	17
0.28	2
5.06	13
9.14	23
11.86	30
11.69	28
6.04	14
7.57	19
1.74	4
9.38	24
0.16	1
1.84	5

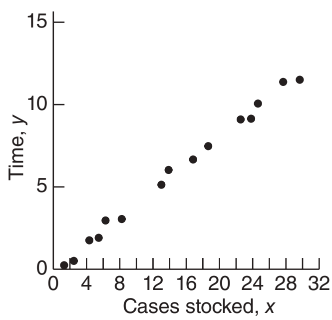


Figure 2.13 Scatter diagram of shelf-stocking data.

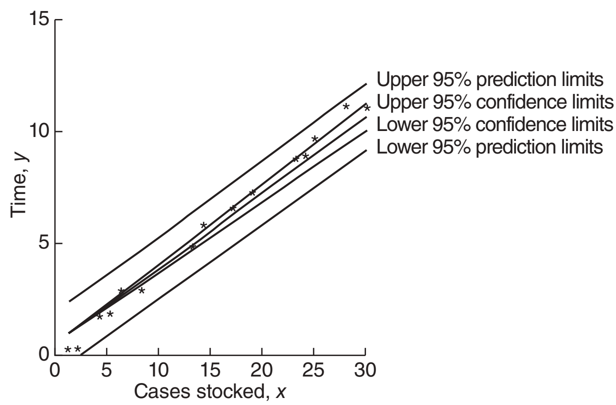


Figure 2.14 The confidence and prediction bands for the shelf-stocing data.

for the intercept model is $MS_{\text{Res}} = 0.0931$ and $R^2 = 0.9997$. Since MS_{Res} for the no-intercept model is smaller than MS_{Res} for the intercept model, we conclude that the no-intercept model is superior. As noted previously, the R^2 statistics are not directly comparable.

Figure 2.14 also shows the 95% confidence interval or $E(y|x_0)$ computed from Eq. (2.54) and the 95% prediction interval on a single future observation y_0 at $x = x_0$ computed from Eq. (2.55). Notice that the length of the confidence interval at $x_0 = 0$ is zero.

SAS handles the no-intercept case. For this situation, the model statement follows:

```
model time = cases/noint
```

2.11 ESTIMATION BY MAXIMUM LIKELIHOOD

The method of least squares can be used to estimate the parameters in a linear regression model regardless of the form of the distribution of the errors ε. Least squares produces best linear unbiased estimators of β_0 and β_1. Other statistical procedures, such as hypothesis testing and CI construction, assume that the errors are normally distributed. If the form of the distribution of the errors is known, an alternative method of parameter estimation, the **method of maximum likelihood**, can be used.

Consider the data $(y_i, x_i), i = 1, 2, \ldots, n$. If we assume that the errors in the regression model are NID(0, σ^2), then the observations y_i in this sample are normally and independently distributed random variables with mean $\beta_0 + \beta_1 x_i$ and variance σ^2. The likelihood function is found from the joint distribution of the observations. If we consider this joint distribution with the observations given and the parameters β_0, β_1, and σ^2 unknown constants, we have the likelihood function. For the simple linear regression model with normal errors, the **likelihood function** is

$$\begin{aligned} L\left(y_i, x_i, \beta_0, \beta_i, \sigma^2\right) &= \prod_{i=1}^{n} \left(2\pi\sigma^2\right)^{-1/2} \exp\left[-\frac{1}{2\sigma^2}(y_i - \beta_0 - \beta_1 x_i)^2\right] \\ &= \left(2\pi\sigma^2\right)^{-1/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2\right] \end{aligned} \tag{2.56}$$

The maximum-likelihood estimators are the parameter values, say $\tilde{\beta}_0$, $\tilde{\beta}_1$, and $\tilde{\sigma}^2$, that maximize L, or equivalently, ln L. Thus,

$$\begin{aligned} \ln L\left(y_i, x_i, \beta_0, \beta_i, \sigma^2\right) &= -\left(\frac{n}{2}\right)\ln 2\pi - \left(\frac{n}{2}\right)\ln \sigma^2 \\ &= -\left(\frac{1}{2\sigma^2}\right)\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2 \end{aligned} \tag{2.57}$$

and the maximum-likelihood estimators $\tilde{\beta}_0$, $\tilde{\beta}_1$, and $\tilde{\sigma}^2$ must satisfy

$$\left.\frac{\partial \ln L}{\partial \beta_0}\right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = \frac{1}{\tilde{\sigma}^2}\sum_{i=1}^{n}\left(y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i\right) = 0 \tag{2.58a}$$

$$\left.\frac{\partial \ln L}{\partial \beta_1}\right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = \frac{1}{\tilde{\sigma}^2}\sum_{i=1}^{n}\left(y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i\right)x_i = 0 \tag{2.58b}$$

and

$$\left.\frac{\partial \ln L}{\partial \sigma^2}\right|_{\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\sigma}^2} = -\frac{n}{2\tilde{\sigma}^2} + \frac{1}{2\tilde{\sigma}^4}\sum_{i=1}^{n}\left(y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i\right)^2 = 0 \tag{2.58c}$$

The solution to Eq. (2.58) gives the **maximum-likelihood estimators:**

$$\tilde{\beta}_0 = \bar{y} - \tilde{\beta}_1 \bar{x} \tag{2.59a}$$

$$\tilde{\beta} = \frac{\sum_{i=1}^{n} y_i (x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{2.59b}$$

$$\tilde{\sigma}^2 = \frac{\sum_{i=1}^{n} \left(y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i\right)^2}{n} \tag{2.59c}$$

Notice that the maximum-likelihood estimators of the intercept and slope, $\tilde{\beta}_0$ and $\tilde{\beta}_1$, are identical to the least-squares estimators of these parameters. Also, $\tilde{\sigma}^2$ is a biased estimator of σ^2. The biased estimator is related to the unbiased estimator $\hat{\sigma}^2$ [Eq. (2.19)] by $\tilde{\sigma}^2 = [(n-1)/n]\hat{\sigma}^2$. The bias is small if n is moderately large. Generally the unbiased estimator $\hat{\sigma}^2$ is used.

In general, maximum-likelihood estimators have better **statistical properties** than least-squares estimators. The maximum-likelihood estimators are **unbiased** (including $\tilde{\sigma}^2$, which is **asymptotically unbiased**, or unbiased as n becomes large) and have **minimum variance** when compared to **all** other unbiased estimators. They are also **consistent estimators** (consistency is a large-sample property indicating that the estimators differ from the true parameter value by a very small amount as n becomes large), and they are a set of **sufficient statistics** (this implies that the estimators contain all of the "information" in the original sample of size n). On the other hand, maximum-likelihood estimation requires more stringent statistical assumptions than the least-squares estimators. The least-squares estimators require only second-moment assumptions (assumptions about the expected value, the variances, and the covariances among the random errors). The maximum-likelihood estimators require a full distributional assumption, in this case that the random errors follow a normal distribution with the same second moments as required for the least-squares estimates. For more information on maximum-likelihood estimation in regression models, see Graybill [1961, 1976], Myers [1990], Searle [1971], and Seber [1977].

2.12 CASE WHERE THE REGRESSOR *x* IS RANDOM

The linear regression model that we have presented in this chapter assumes that the values of the regressor variable x are known constants. This assumption makes the confidence coefficients and type I (or type II) errors refer to repeated sampling on y at the same x levels. There are many situations in which assuming that the x's are fixed constants is inappropriate. For example, consider the soft drink delivery time data from Chapter 1 (Figure 1.1). Since the outlets visited by the delivery person are selected at random, it is unrealistic to believe that we can control the delivery volume x. It is more reasonable to assume that both y and x are random variables.

Fortunately, under certain circumstances, all of our earlier results on parameter estimation, testing, and prediction are valid. We now discuss these situations.

2.12.1 *x* and *y* Jointly Distributed

Suppose that x and y are jointly distributed random variables but the form of this joint distribution is unknown. It can be shown that all of our previous regression results hold if the following conditions are satisfied:

1. The conditional distribution of y given x is normal with conditional mean $\beta_0 + \beta_1 x$ and conditional variance σ^2.
2. The x's are independent random variables whose probability distribution does not involve β_0, β_1, and σ^2.

While all of the regression procedures are unchanged when these conditions hold, the confidence coefficients and statistical errors have a different interpretation. When the regressor is a random variable, these quantities apply to repeated sampling of (x_i, y_i) values and not to repeated sampling of y_i at fixed levels of x_i.

2.12.2 *x* and *y* Jointly Normally Distributed: Correlation Model

Now suppose that y and x are jointly distributed according to the **bivariate normal distribution**. That is,

$$f(y, x) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{y-\mu_1}{\sigma_1}\right)^2 + \left(\frac{x-\mu_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{y-\mu_1}{\sigma_1}\right)\left(\frac{x-\mu_2}{\sigma_2}\right)\right]\right\} \tag{2.60}$$

where μ_1 and σ_1^2 the mean and variance of y, μ_2 and σ_2^2 the mean and variance of x, and

$$\rho = \frac{E(y-\mu_1)(x-\mu_2)}{\sigma_1\sigma_2} = \frac{\sigma_{12}}{\sigma_1\sigma_2}$$

is the **correlation coefficient** between y and x. The term σ_{12} is the **covariance** of y and x.

The **conditional distribntion** of y for a given value of x is

$$f(y|x) = \frac{1}{\sqrt{2\pi}\sigma_{1.2}} \exp\left[-\frac{1}{2}\left(\frac{y-\beta_0-\beta_1 x}{\sigma_{1.2}}\right)^2\right] \tag{2.61}$$

where

$$\beta_0 = \mu_1 - \mu_2\rho\frac{\sigma_1}{\sigma_2} \tag{2.62a}$$

$$\beta_1 = \frac{\sigma_1}{\sigma_2}\rho \tag{2.62b}$$

and

$$\sigma_{1.2}^2 = \sigma_1^2(1-\rho^2) \tag{2.62c}$$

That is, the conditional distribution of y given x is normal with conditional mean

$$E(y|x) = \beta_0 + \beta_1 x \tag{2.63}$$

and conditional variance $\sigma_{1.2}^2$. Note that the mean of the conditional distribution of y given x is a straight-line regression model. Furthermore, there is a relationship between the correlation coefficient ρ and the slope β_1. From Eq. (2.62b) we see that if $\rho = 0$, then $\beta_1 = 0$, which implies that there is no linear regression of y on x. That is, knowledge of x does not assist us in predicting y.

The method of maximum likelihood may be used to estimate the parameters β_0 and β_1. It may be shown that the maximum-likelihood estimators of these parameters are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{2.64a}$$

and

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} \tag{2.64b}$$

The estimators of the intercept and slope in Eq. (2.64) are identical to those given by the method of least squares in the case where x was assumed to be a controllable variable. In general, the regression model with y and x jointly normally distributed may be analyzed by the methods presented previously for the model with x a controllable variable. This follows because the random variable y given x is independently and normally distributed with mean $\beta_0 + \beta_1 x$ and constant variance $\sigma_{1.2}^2$. As noted in Section 2.12.1, these results will also hold for **any** joint distribution of y and x such that the conditional distribution of y given x is normal.

It is possible to draw inferences about the correlation coefficient ρ in this model. The estimator of ρ is the **sample correlation coefficient**

$$r = \frac{\sum_{i=1}^{n} y_i(x_i - \bar{x})}{\left[\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2\right]^{1/2}} = \frac{S_{xy}}{[S_{xx} SS_{\mathrm{T}}]^{1/2}} \tag{2.65}$$

Note that

$$\hat{\beta}_1 = \left(\frac{SS_{\mathrm{T}}}{S_{xx}}\right)^{1/2} r \tag{2.66}$$

so that the slope $\hat{\beta}$ is just the sample correlation coefficient r multiplied by a scale factor that is the square root of the spread of the y's divided by the spread of the x's. Thus, $\hat{\beta}_1$ and r are closely related, although they provide somewhat different information. The sample correlation coefficient r is a measure of the **linear association** between y and x, while $\hat{\beta}_1$ measures the change in the mean of y for a unit change in x. In the case of a controllable variable x, r has no meaning because the magnitude of r depends on the choice of spacing for x. We may also write, from Eq. (2.66),

$$r^2 = \hat{\beta}_1^2 \frac{S_{xx}}{SS_T} = \frac{\hat{\beta}_1 S_{xy}}{SS_T} = \frac{SS_R}{SS_T} = R^2$$

which we recognize from Eq. (2.47) as the coefficient of determination. That is, the coefficient of determination R^2 is just the square of the correlation coefficient between y and x.

While regression and correlation are closely related, regression is a more powerful tool in many situations. Correlation is only a measure of association and is of little use in prediction. However, regression methods are useful in developing quantitative relationships between variables, which can be used in prediction.

It is often useful to test the hypothesis that the correlation coefficient equals zero, that is,

$$H_0\colon \rho = 0, \quad H_1\colon \rho \neq 0 \tag{2.67}$$

The appropriate test statistic for this hypothesis is

$$t_0 = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \tag{2.68}$$

which follows the t distribution with $n - 2$ degrees of freedom if $H_0\colon \rho = 0$ is true. Therefore, we would reject the null hypothesis if $|t_0| > t_{\alpha/2,\, n-2}$. This test is equivalent to the t test for $H_0\colon \beta_1 = 0$ given in Section 2.3. This equivalence follows directly from Eq. (2.66).

The test procedure for the hypotheses

$$H_0\colon \rho = \rho_0, \quad H_1\colon \rho \neq \rho_0 \tag{2.69}$$

where $\rho_0 \neq 0$ is somewhat more complicated. For moderately large samples (e.g., $n \geq 25$) the statistic

$$Z = \operatorname{arctanh} r = \frac{1}{2}\ln\frac{1+r}{1-r} \tag{2.70}$$

is approximately normally distributed with mean

$$\mu_Z = \operatorname{arctanh} \rho = \frac{1}{2}\ln\frac{1+\rho}{1-\rho}$$

and variance

$$\sigma_Z^2 = (n-3)^{-1}$$

Therefore, to test the hypothesis H_0: $\rho = \rho_0$, we may compute the statistic

$$Z_0 = (\operatorname{arctanh} r - \operatorname{arctanh} \rho_0)(n-3)^{1/2} \tag{2.71}$$

and reject H_0: $\rho = \rho_0$ if $|Z_0| > Z_{\alpha/2}$.

It is also possible to construct a 100(1 − α) percent CI for ρ using the transformation (2.70). The 100(1 − α) percent CI is

$$\tanh\left(\operatorname{arctanh} r - \frac{Z_{\alpha/2}}{\sqrt{n-3}}\right) \le \rho \le \tanh\left(\operatorname{arctanh} r + \frac{Z_{\alpha/2}}{\sqrt{n-3}}\right) \tag{2.72}$$

where $\tanh u = (e^u - e^{-u})/(e^u + e^{-u})$.

Example 2.9 The Delivery Time Data

Consider the soft drink delivery time data introduced in Chapter 1. The 25 observations on delivery time y and delivery volume x are listed in Table 2.11. The scatter diagram shown in Figure 1.1 indicates a strong linear relationship between delivery time and delivery volume. The Minitab output for the simple linear regression model is in Table 2.12.

The sample correlation coefficient between delivery time y and delivery volume x is

$$r = \frac{S_{xy}}{[S_{xx}SS_{\mathrm{T}}]^{1/2}} = \frac{2473.3440}{[(1136.5600)(5784.5426)]^{1/2}} = 0.9646$$

TABLE 2.11 Data Example 2.9

Observation	Delivery Time, y	Number of Cases, x	Observation	Delivery Time, y	Number of Cases, x
1	16.68	7	14	19.75	6
2	11.50	3	15	24.00	9
3	12.03	3	16	29.00	10
4	14.88	4	17	15.35	6
5	13.75	6	18	19.00	7
6	18.11	7	19	9.50	3
7	8.00	2	20	35.10	17
8	17.83	7	21	17.90	10
9	79.24	30	22	52.32	26
10	21.50	5	23	18.75	9
11	40.33	16	24	19.83	8
12	21.00	10	25	10.75	4
13	13.50	4			

TABLE 2.12 MINITAB Output for Soft Drink Delivery Time Data

```
Regression Analysis: Time versus Cases

The regression equation is
Time = 3.32 + 2.18 Cases

Predictor        Coef            SE Coef      T        P
Constant         3.321           1.371         2.42    0.024
Cases            2.1762          0.1240       17.55    0.000

S = 4.18140      R- Sq= 93.0%    R- Sq(adj) = 92.7%

Analysis of Variance

Source            DF             SS           MS       F        P
Regression         1             5382.4       5382.4   307.85   0.000
Residual Error    23              402.1         17.5
Total             24             5784.5
```

If we assume that delivery time and delivery volume are jointly normally distributed, we may test the hypotheses

$$H_0\colon \rho = 0, \quad H_1\colon \rho \neq 0$$

using the test statistic

$$t_0 = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.9646\sqrt{23}}{\sqrt{1-0.9305}} = 17.55$$

Since $t_{0.025,23} = 2.069$, we reject H_0 and conclude that the correlation coefficient $\rho \neq 0$. Note from the Minitab output in Table 2.12 that this is identical to the t-test statistic for $H_0\colon \beta_1 = 0$. Finally, we may construct an approximate 95% CI on ρ from (2.72). Since arctanh r = arctanh 0.9646 = 2.0082, Eq. (2.72) becomes

$$\tanh\left(2.0082 - \frac{1.96}{\sqrt{22}}\right) \leq \rho \leq \tanh\left(2.0082 + \frac{1.96}{\sqrt{22}}\right)$$

which reduces to

$$0.9202 \leq \rho \leq 0.9845$$ ■

Although we know that delivery time and delivery volume are highly correlated, this information is of little use in predicting, for example, delivery time as a function of the number of cases of product delivered. This would require a regression model. The straight-line fit (shown graphically in Figure 1.1b) relating delivery time to delivery volume is

$$\hat{y} = 3.321 + 2.1762x$$

Further analysis would be required to determine if this equation is an adequate fit to the data and if it is likely to be a successful predictor.

PROBLEMS

2.1 Table B.1 gives data concerning the performance of the 26 National Football League teams in 1976. It is suspected that the number of yards gained rushing by opponents (x_8) has an effect on the number of games won by a team (y).

a. Fit a simple linear regression model relating games won y to yards gained rushing by opponents x_8.

b. Construct the analysis-of-variance table and test for significance of regression.

c. Find a 95% CI on the slope.

d. What percent of the total variability in y is explained by this model?

e. Find a 95% CI on the mean number of games won if opponents' yards rushing is limited to 2000 yards.

2.2 Suppose we would like to use the model developed in Problem 2.1 to predict the number of games a team will win if it can limit opponents' yards rushing to 1800 yards. Find a point estimate of the number of games won when $x_8 = 1800$. Find a 90% prediction interval on the number of games won.

2.3 Table B.2 presents data collected during a solar energy project at Georgia Tech.

a. Fit a simple linear regression model relating total heat flux y (kilowatts) to the radial deflection of the deflected rays x_4 (milliradians).

b. Construct the analysis-of-variance table and test for significance of regression.

c. Find a 99% CI on the slope.

d. Calculate R^2.

e. Find a 95% CI on the mean heat flux when the radial deflection is 16.5 milliradians.

2.4 Table B.3 presents data on the gasoline mileage performance of 32 different automobiles.

a. Fit a simple linear regression model relating gasoline mileage y (miles per gallon) to engine displacement x_1 (cubic inches).

b. Construct the analysis-of-variance table and test for significance of regression.

c. What percent of the total variability in gasoline mileage is accounted for by the linear relationship with engine displacement?

d. Find a 95% CI on the mean gasoline mileage if the engine displacement is 275 in.^{3}

e. Suppose that we wish to predict the gasoline mileage obtained from a car with a 275-in.^{3} engine. Give a point estimate of mileage. Find a 95% prediction interval on the mileage.

f. Compare the two intervals obtained in parts d and e. Explain the difference between them. Which one is wider, and why?

2.5 Consider the gasoline mileage data in Table B.3. Repeat Problem 2.4 (parts a, b, and c) using vehicle weight x_{10} as the regressor variable. Based on a comparison of the two models, can you conclude that x_1 is a better choice of regressor than x_{10}?

2.6 Table B.4 presents data for 27 houses sold in Erie, Pennsylvania.

a. Fit a simple linear regression model relating selling price of the house to the current taxes (x_1).

b. Test for significance of regression.

c. What percent of the total variability in selling price is explained by this model?

d. Find a 95% CI on β_1.

e. Find a 95% CI on the mean selling price of a house for which the current taxes are $750.

2.7 The purity of oxygen produced by a fractional distillation process is thought to be related to the percentage of hydrocarbons in the main condensor of the processing unit. Twenty samples are shown below.

Purily (%)	Hydrocarbon (%)	Purily (%)	Hydrocarbon (%)
86.91	1.02	96.73	1.46
89.85	1.11	99.42	1.55
90.28	1.43	98.66	1.55
86.34	1.11	96.07	1.55
92.58	1.01	93.65	1.40
87.33	0.95	87.31	1.15
86.29	1.11	95.00	1.01
91.86	0.87	96.85	0.99
95.61	1.43	85.20	0.95
89.86	1.02	90.56	0.98

a. Fit a simple linear regression model to the data.

b. Test the hypothesis H_0: $\beta_1 = 0$.

c. Calculate R^2.

d. Find a 95% CI on the slope.

e. Find a 95% CI on the mean purity when the hydrocarbon percentage is 1.00.

2.8 Consider the oxygen plant data in Problem 2.7 and assume that purity and hydrocarbon percentage are jointly normally distributed random variables.

a. What is the correlation between oxygen purity and hydrocarbon percentage?

b. Test the hypothesis that $\rho = 0$.

c. Construct a 95% CI for ρ.

2.9 Consider the soft drink delivery time data in Table 2.9. After examining the original regression model (Example 2.9), one analyst claimed that the model was invalid because the intercept was not zero. He argued that if zero cases were delivered, the time to stock and service the machine would be zero, and the straight-line model should go through the origin. What would you say in response to his comments? Fit a no-intercept model to these data and determine which model is superior.

2.10 The weight and systolic blood pressure of 26 randomly selected males in the age group 25–30 are shown below. Assume that weight and blood pressure (BP) are jointly normally distributed.

a. Find a regression line relating systolic blood pressure to weight.

b. Estimate the correlatiou coefficient.

c. Test the hypothesis that $\rho = 0$.

d. Test the hypothesis that $\rho = 0.6$.

e. Find a 95% CI for ρ.

Subject	Weight	Symbolic BP	Subject	Weight	Systolic BP
1	165	130	14	172	153
2	167	133	15	159	128
3	180	150	16	168	132
4	155	128	17	174	149
5	212	151	18	183	158
6	175	146	19	215	150
7	190	150	20	195	163
8	210	140	21	180	156
9	200	148	22	143	124
10	149	125	23	240	170
11	158	133	24	235	165
12	169	135	25	192	160
13	170	150	26	187	159

2.11 Consider the weight and blood pressure data in Problem 2.10. Fit a no-intercept model to the data and compare it to the model obtained in Problem 2.10. Which model would you conclude is superior?

2.12 The number of pounds of steam used per month at a plant is thought to be related to the average monthly ambient temperature. The past year's usages and temperatures follow.

Month	Temperature	Usage/l000	Month	Temperature	Usage/l000
Jan.	21	185.79	Jul.	68	621.55
Feb.	24	214.47	Aug.	74	675.06
Mar.	32	288.03	Sep.	62	562.03
Apr.	47	424.84	Oct.	50	452.93
May	50	454.68	Nov.	41	369.95
Jun.	59	539.03	Dec.	30	273.98

a. Fit a simple linear regression model to the data.

b. Test for significance of regression.

c. Plant management believes that an increase in average ambient temperature of 1 degree will increase average monthly steam consumption by 10,000 lb. Do the data support this statement?

d. Construct a 99% prediction interval on steam usage in a month with average ambient temperature of 58°.

2.13 Davidson ("Update on Ozone Trends in California's South Coast Air Basin," *Air and Waste*, **43**, 226, 1993) studied the ozone levels in the South Coast Air Basin of California for the years 1976–1991. He believes that the number of days the ozone levels exceeded 0.20 ppm (the response) depends on the seasonal meteorological index, which is the seasonal average 850-millibar temperature (the regressor). The following table gives the data.

Year	Days	Index
1976	91	16.7
1977	105	17.1
1978	106	18.2
1979	108	18.1
1980	88	17.2
1981	91	18.2
1982	58	16.0
1983	82	17.2
1984	81	18.0
1985	65	17.2
1986	61	16.9
1987	48	17.1
1988	61	18.2
1989	43	17.3
1990	33	17.5
1991	36	16.6

a. Make a scatterplot of the data.

b. Estimate the prediction equation.

c. Test for significance of regression.

d. Calculate and plot the 95% confidence and prediction bands.

2.14 Hsuie, Ma, and Tsai ("Separation and Characterizations of Thermotropic Copolyesters of *p*-Hydroxybenzoic Acid, Sebacic Acid, and Hydroquinone," *Journal of Applied Polymer Science*, **56**, 471–476, 1995) study the effect of the molar ratio of sebacic acid (the regressor) on the intrinsic viscosity of copolyesters (the response). The following table gives the data.

Ratio	Viscosity
1.0	0.45
0.9	0.20
0.8	0.34
0.7	0.58
0.6	0.70
0.5	0.57
0.4	0.55
0.3	0.44

a. Make a scatterplot of the data.

b. Estimate the prediction equation.

c. Perform a complete, appropriate analysis (statistical tests, calculation of R^2, and so forth).

d. Calculate and plot the 95% confidence and prediction bands.

2.15 Byers and Williams ("Viscosities of Binary and Ternary Mixtures of Polynomatic Hydrocarbons," *Journal of Chemical and Engineering Data*, **32**, 349–354, 1987) studied the impact of temperature on the viscosity of toluene–tetralin blends. The following table gives the data for blends with a 0.4 molar fraction of toluene.

Temperature (°C)	Viscosity (mPa · s)
24.9	1.1330
35.0	0.9772
44.9	0.8532
55.1	0.7550
65.2	0.6723
75.2	0.6021
85.2	0.5420
95.2	0.5074

a. Estimate the prediction equation.

b. Perform a complete analysis of the model.

c. Calculate and plot the 95% confidence and prediction bands.

2.16 Carroll and Spiegelman ("The Effects of Ignoring Small Measurement Errors in Precision Instrument Calibration," *Journal of Quality Technology*, **18**, 170–173, 1986) look at the relationship between the pressure in a tank and the volume of liquid. The following table gives the data. Use an appropriate statistical software package to perform an analysis of these data. Comment on the output produced by the software routine.

Volume	Pressure	Volume	Pressure	Volume	Pressure
2084	4599	2842	6380	3789	8599
2084	4600	3030	6818	3789	8600
2273	5044	3031	6817	3979	9048
2273	5043	3031	6818	3979	9048
2273	5044	3221	7266	4167	9484
2463	5488	3221	7268	4168	9487
2463	5487	3409	7709	4168	9487
2651	5931	3410	7710	4358	9936
2652	5932	3600	8156	4358	9938
2652	5932	3600	8158	4546	10377
2842	6380	3788	8597	4547	10379

2.17 Atkinson (*Plots, Transformations, and Regression*, Clarendon Press, Oxford, 1985) presents the following data on the boiling point of water (°F) and barometric pressure (inches of mercury). Construct a scatterplot of the data and propose a model that relates a model that relates boiling point to barometric pressure. Fit the model to the data and perform a complete analysis of the model using the techniques we have discussed in this chapter.

Boiling Point	Barometric Pressure	Boiling Point	Barometric Pressure
199.5	20.79	201.9	24.02
199.3	20.79	201.3	24.01
197.9	22.40	203.6	25.14
198.4	22.67	204.6	26.57
199.4	23.15	209.5	28.49
199.9	23.35	208.6	27.76
200.9	23.89	210.7	29.64
201.1	23.99	211.9	29.88
		212.2	30.06

2.18 On March 1, 1984, the *Wall Street Journal* published a survey of television advertisements conducted by Video Board Tests, Inc., a New York ad-testing company that interviewed 4000 adults. These people were regular product users who were asked to cite a commercial they had seen for that product category in the past week. In this case, the response is the number of millions of retained impressions per week. The regressor is the amount of money spent by the firm on advertising. The data follow.

Firm	Amount Spent (millions)	Returned Impressions per week (millions)
Miller Lite	50.1	32.1
Pepsi	74.1	99.6
Stroh's	19.3	11.7

(*Continued*)

Firm	Amount Spent (millions)	Returned Impressions per week (millions)
Federal Express	22.9	21.9
Burger King	82.4	60.8
Coca-Cola	40.1	78.6
McDonald's	185.9	92.4
MCI	26.9	50.7
Diet Cola	20.4	21.4
Ford	166.2	40.1
Levi's	27	40.8
Bud Lite	45.6	10.4
ATT Bell	154.9	88.9
Calvin Klein	5	12
Wendy's	49.7	29.2
Polaroid	26.9	38
Shasta	5.7	10
Meow Mix	7.6	12.3
Oscar Meyer	9.2	23.4
Crest	32.4	71.1
Kibbles N Bits	6.1	4.4

a. Fit the simple linear regression model to these data.

b. Is there a significant relationship between the amount a company spends on advertising and retained impressions? Justify your answer statistically.

c. Construct the 95% confidence and prediction bands for these data.

d. Give the 95% confidence and prediction intervals for the number of retained impressions for MCI.

2.19 Table B.17 Contains the Patient Satisfaction data used in Section 2.7.

a. Fit a simple linear regression model relating satisfaction to age.

b. Compare this model to the fit in Section 2.7 relating patient satisfaction to severity.

2.20 Consider the fuel consumption data given in Table B.18. The automotive engineer believes that the initial boiling point of the fuel controls the fuel consumption. Perform a thorough analysis of these data. Do the data support the engineer's belief?

2.21 Consider the wine quality of young red wines data in Table B.19. The winemakers believe that the sulfur content has a negative impact on the taste (thus, the overall quality) of the wine. Perform a thorough analysis of these data. Do the data support the winemakers' belief?

2.22 Consider the methanol oxidation data in Table B.20. The chemist believes that ratio of inlet oxygen to the inlet methanol controls the conversion process. Perform a through analysis of these data. Do the data support the chemist's belief?

2.23 Consider the simple linear regression model $y = 50 + 10x + \varepsilon$ where ε is NID (0, 16). Suppose that $n = 20$ pairs of observations are used to fit this model.

Generate 500 samples of 20 observations, drawing one observation for each level of $x = 1, 1.5, 2, \ldots, 10$ for each sample.

a. For each sample compute the least-squares estimates of the slope and intercept. Construct histograms of the sample values of $\hat{\beta}_0$ and $\hat{\beta}_1$. Discuss the shape of these histograms.

b. For each sample, compute an estimate of $E(y|x = 5)$. Construct a histogram of the estimates you obtained. Discuss the shape of the histogram.

c. For each sample, compute a 95% CI on the slope. How many of these intervals contain the true value $\beta_1 = 10$? Is this what you would expect?

d. For each estimate of $E(y|x = 5)$ in part b, compute the 95% CI. How many of these intervals contain the true value of $E(y|x = 5) = 100$? Is this what you would expect?

2.24 Repeat Problem 2.20 using only 10 observations in each samle, drawing one observation from each level $x = 1, 2, 3, \ldots, 10$. What impact does using $n = 10$ have on the questions asked in Problem 2.17? Compare the lengths of the CIs and the appearance of the histograms.

2.25 Consider the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$, with $E(\varepsilon) = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2$, and ε uncorrelated.

a. Show that $\mathrm{Cov}\left(\hat{\beta}_0, \hat{\beta}_1\right) = -\bar{x}\sigma^2/S_{xx}$.

b. Show that $\mathrm{Cov}\left(\bar{y}, \beta_1\right) = 0$.

2.26 Consider the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$, with $E(\varepsilon) = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2$, and ε uncorrelated.

a. Show that $E(MS_{\mathrm{R}}) = \sigma^2 + \beta_1^2 S_{xx}$.

b. Show that $E(MS_{\mathrm{Res}}) = \sigma^2$.

2.27 Suppose that we have fit the straight-line regression model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$ but the response is affected by a second variable x_2 such that the true regression function is

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

a. Is the least-squares estimator of the slope in the original simple linear regression model unbiased?

b. Show the bias in $\hat{\beta}_1$.

2.28 Consider the maximum-likelihood estimator $\tilde{\sigma}^2$ of σ^2 in the simple linear regression model. We know that $\tilde{\sigma}^2$ is a biased estimator for σ^2.

a. Show the amount of bias in $\tilde{\sigma}^2$.

b. What happens to the bias as the sample size n becomes large?

2.29 Suppose that we are fitting a straight line and wish to make the standard error of the slope as small as possible. Suppose that the "region of interest" for x is $-1 \leq x \leq 1$. Where should the observations $x_1, x_2, \ldots, x_n$ be taken? Discuss the practical aspects of this data collection plan.

2.30 Consider the data in Problem 2.12 and assume that steam usage and average temperature are jointly normally distributed.

a. Find the correlation between steam usage and monthly average ambient temperature.

b. Test the hypothesis that $\rho = 0$.

c. Test the hypothesis that $\rho = 0.5$.

d. Find a 99% CI for ρ.

2.31 Prove that the maximum value of R^2 is less than 1 if the data contain repeated (different) observations on y at the same value of x.

2.32 Consider the simple linear regression model

$$y = \beta_0 + \beta_1 x + \varepsilon$$

where the intercept β_0 is known.

a. Find the least-squares estimator of β_1 for this model. Does this answer seem reasonable?

b. What is the variance of the slope $\left(\hat{\beta}_1\right)$ for the least-squares estimator found in part a?

c. Find a $100(1 - \alpha)$ percent CI for β_1. Is this interval narrower than the estimator for the case where both slope and intercept are unknown?

2.33 Consider the least-squares residuals $e_i = y_i - \hat{y}_i, i = 1, 2, \ldots, n$, from the simple linear regression model. Find the variance of the residuals $\mathrm{Var}(e_i)$. Is the variance of the residuals a constant? Discuss.